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(54) **VEHICULAR ANTENNA DEVICE**

(56) **References Cited**

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(57) **ABSTRACT**

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H01Q 5/307 (2015.01)

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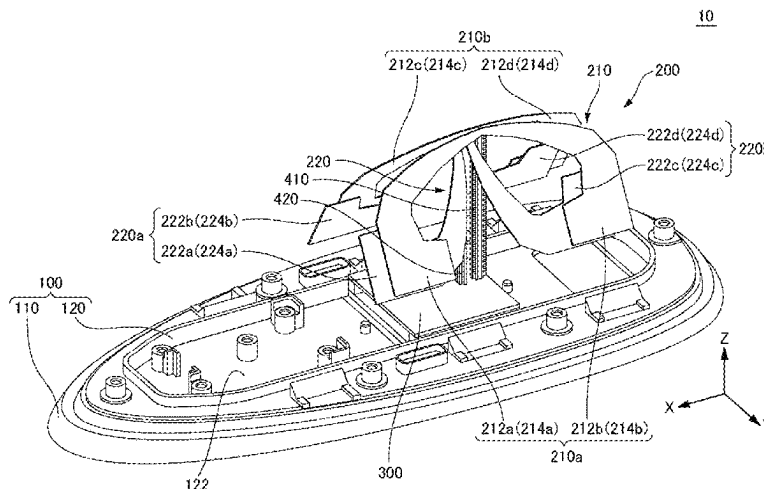
(52) **U.S. Cl.**
CPC **H01Q 1/3275** (2013.01); **H01Q 5/307**
(2015.01); **H01Q 9/28** (2013.01); **H01Q 21/24**
(2013.01); **H01Q 21/26** (2013.01)

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CPC H01Q 21/26; H01Q 1/32; H01Q 1/325;
H01Q 1/3275

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An antenna element (200) includes a first element (210) having a first feed portion, and a first element section (210a) and a second element section (210b) at opposite sides of the first feed portion, and a second element (220) having a second feed portion and, a third element section (220a) and a fourth element section (220b) at opposite sides of the second feed portion, at least a part of the first element (210) and at least a part of the second element (220) face each other, and one of the first element section (210a) and the second element section (210b) is arranged at an angle with respect to the other of the first element section (210a) and the second element section (210b).

13 Claims, 17 Drawing Sheets



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FIG. 1

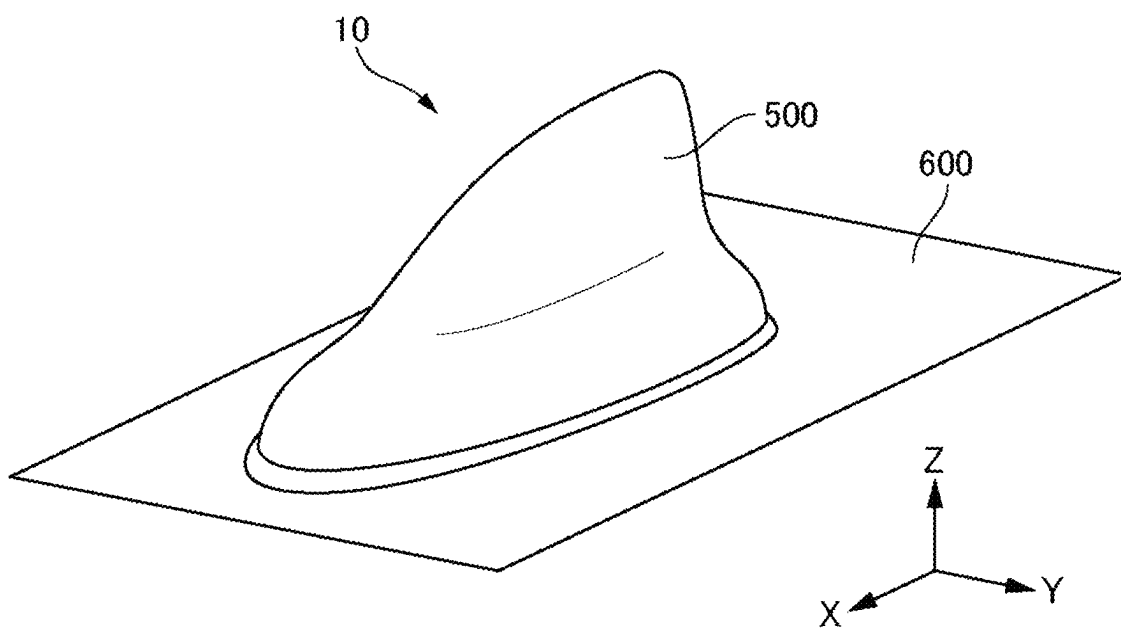


FIG. 2

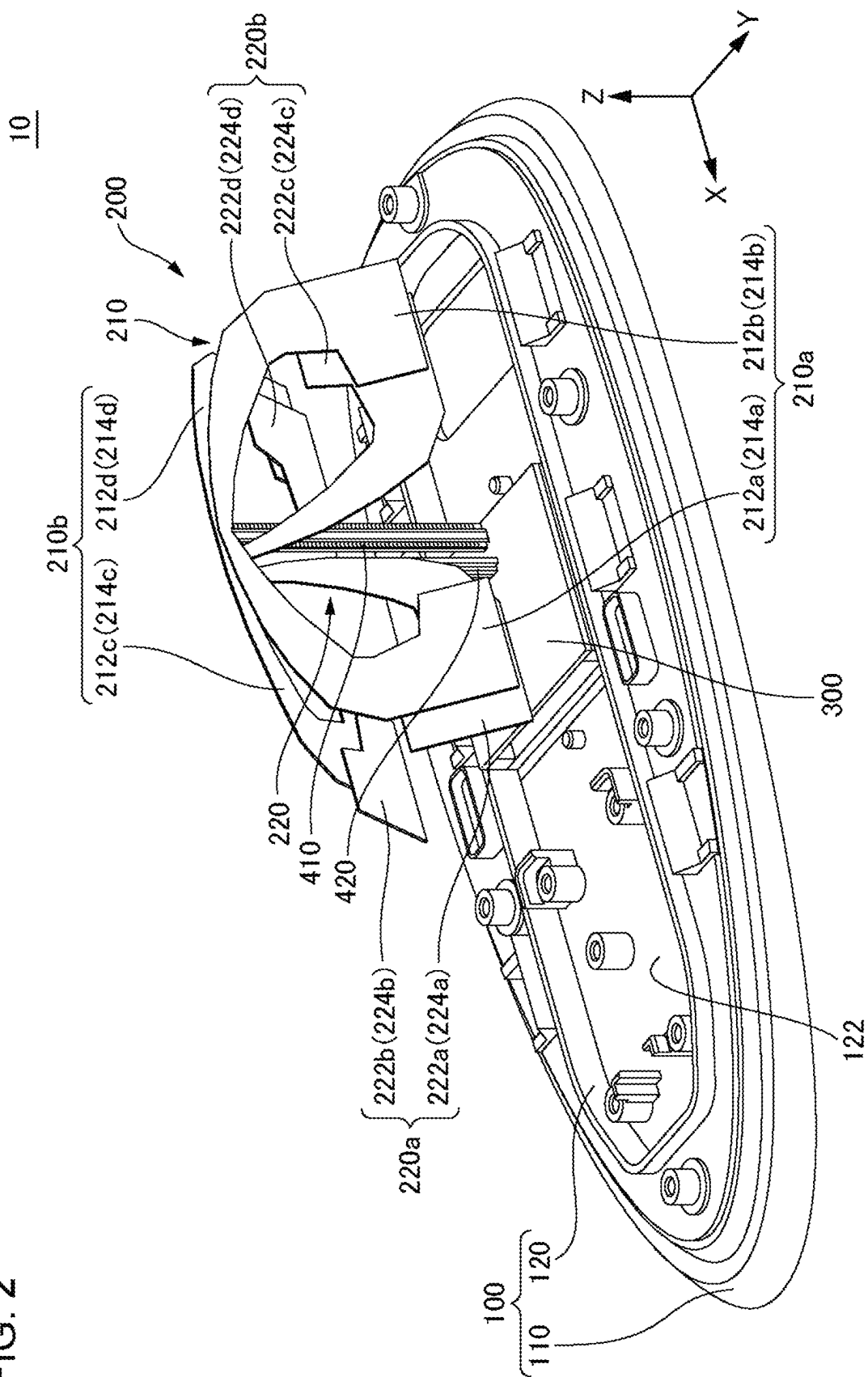


FIG. 3

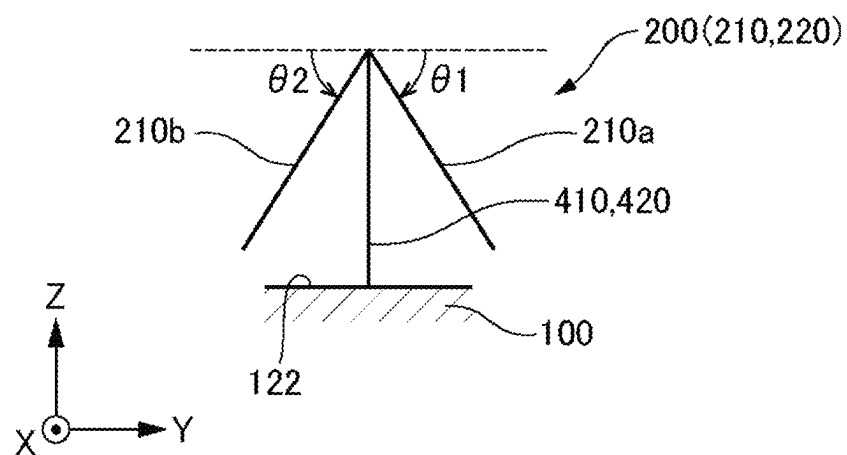


FIG. 4

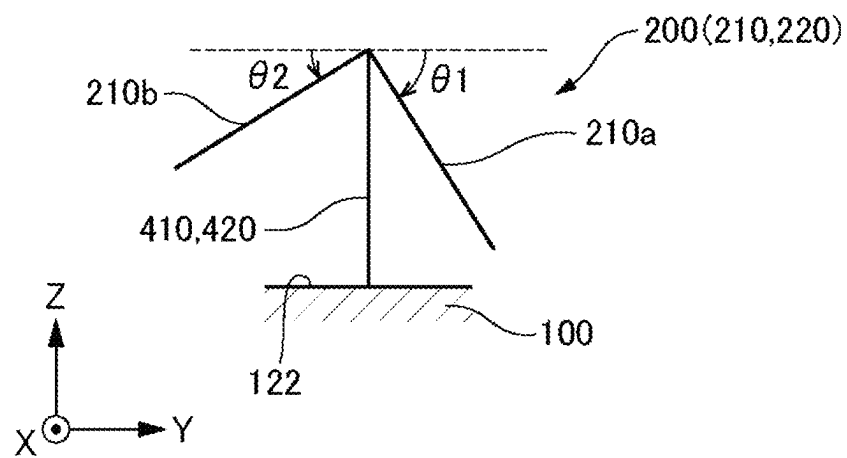


FIG. 5

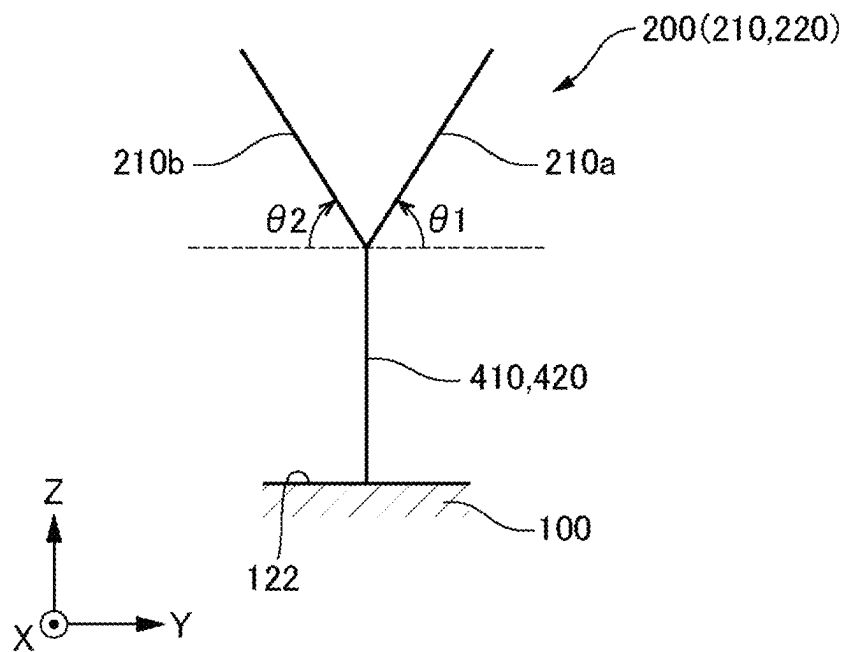


FIG. 6

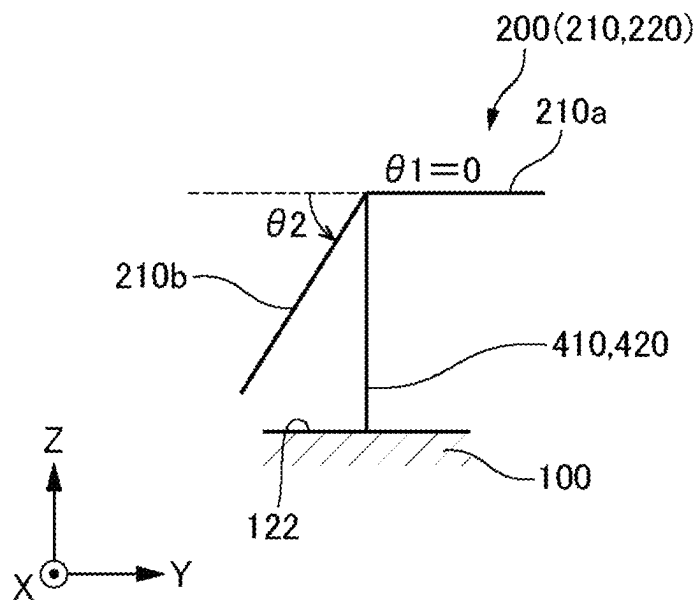


FIG. 7

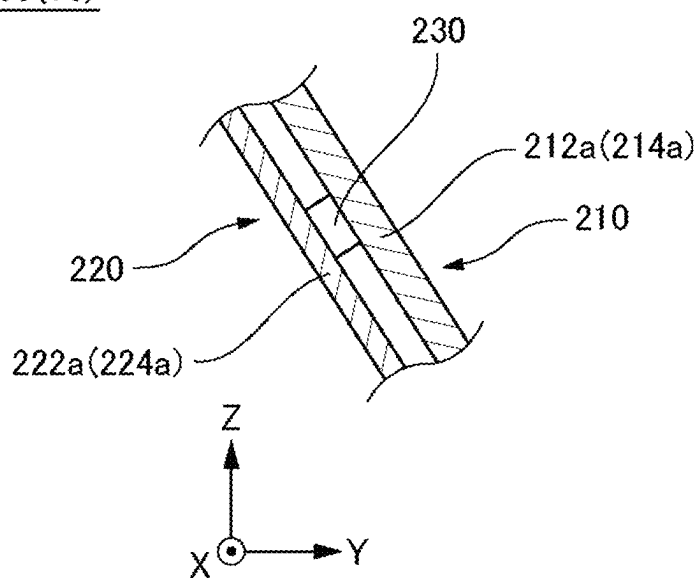
200(10)

FIG. 8

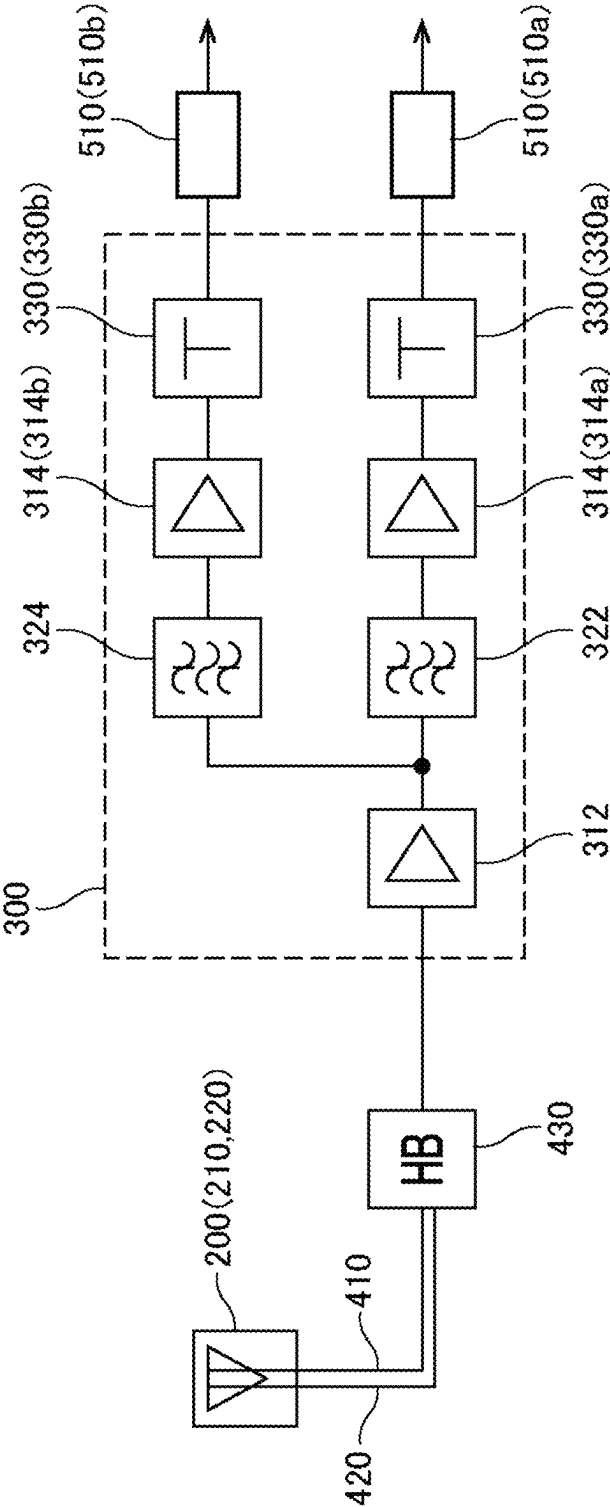


FIG. 9

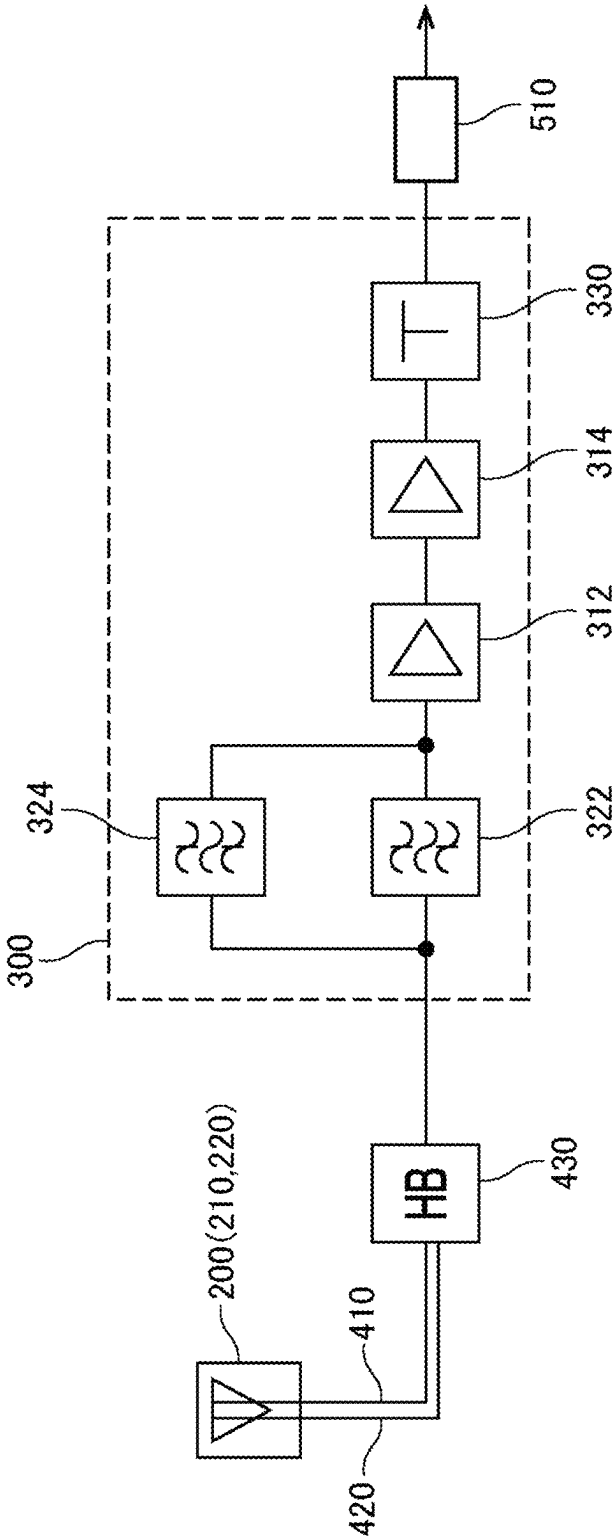


FIG. 10

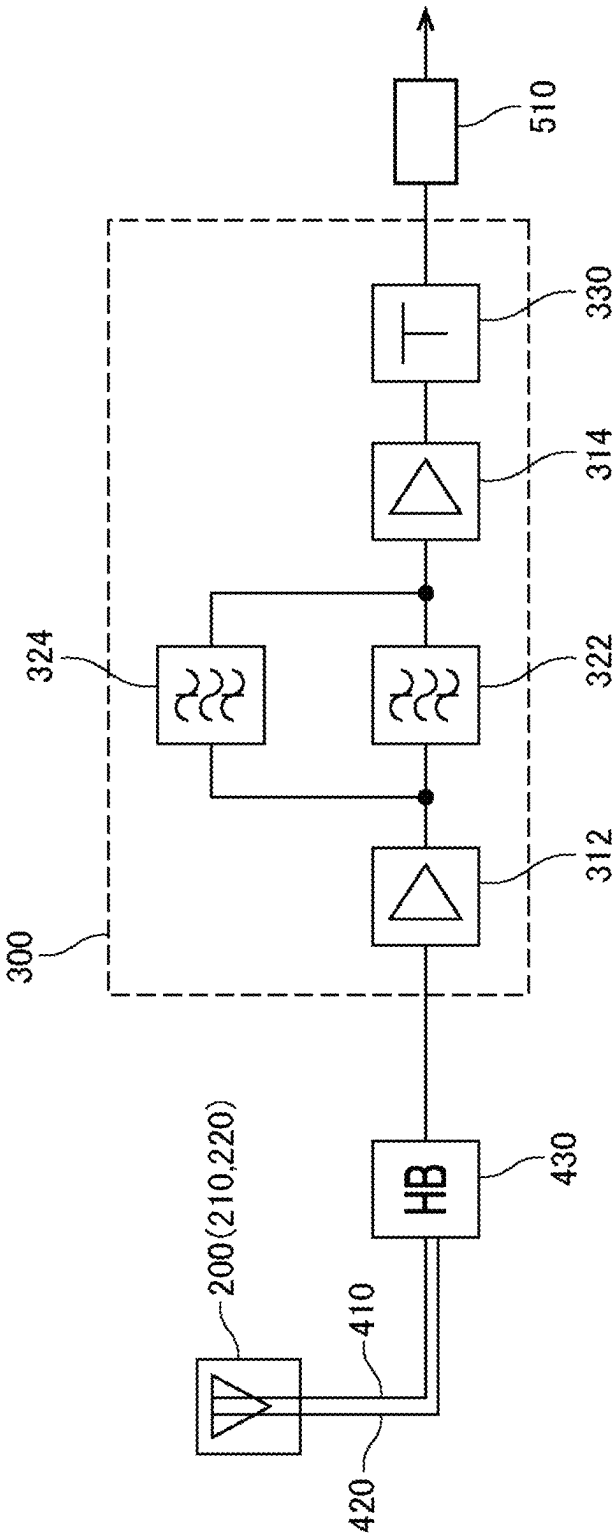


FIG. 11

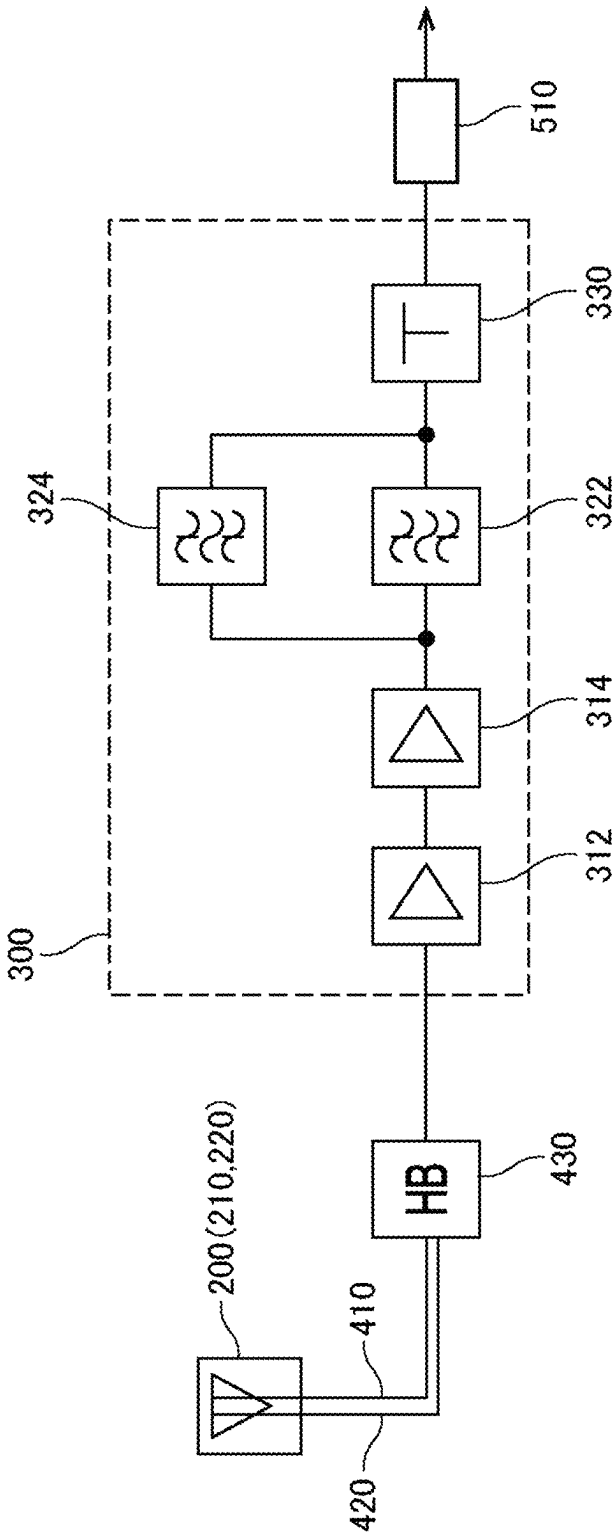


FIG. 12

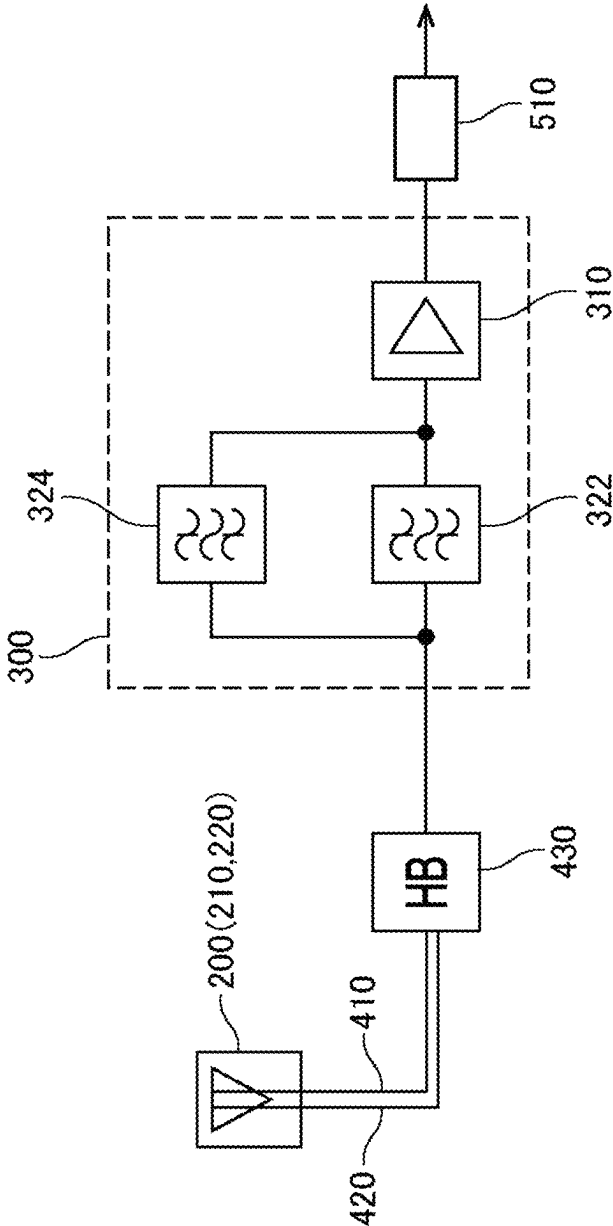


FIG. 13

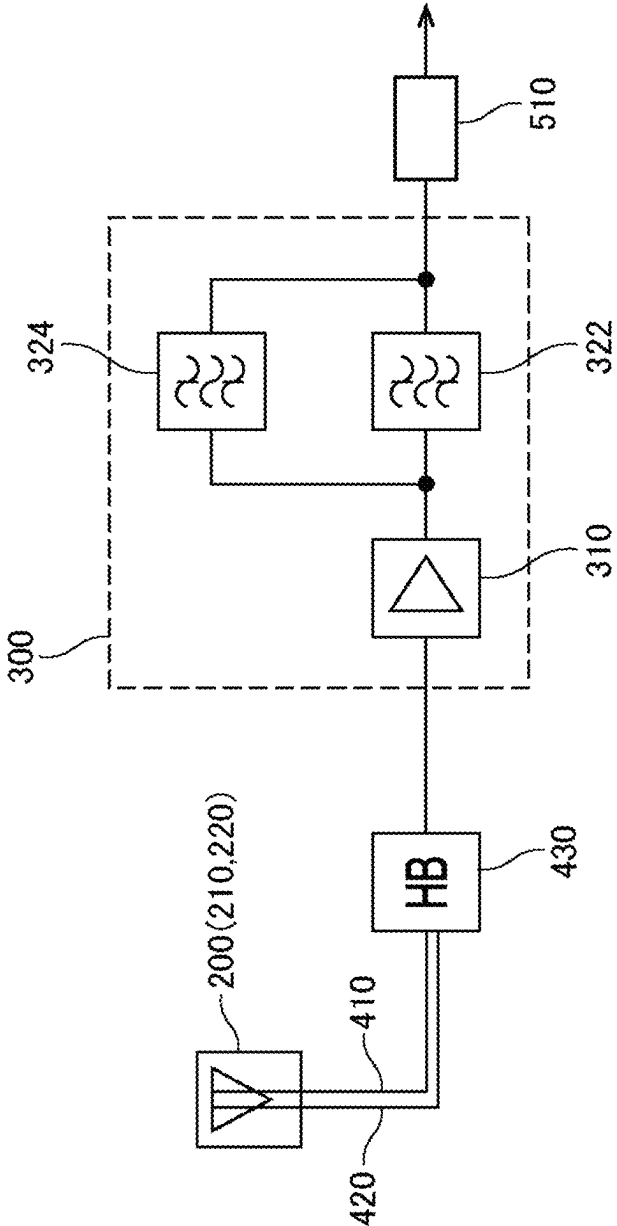


FIG. 14

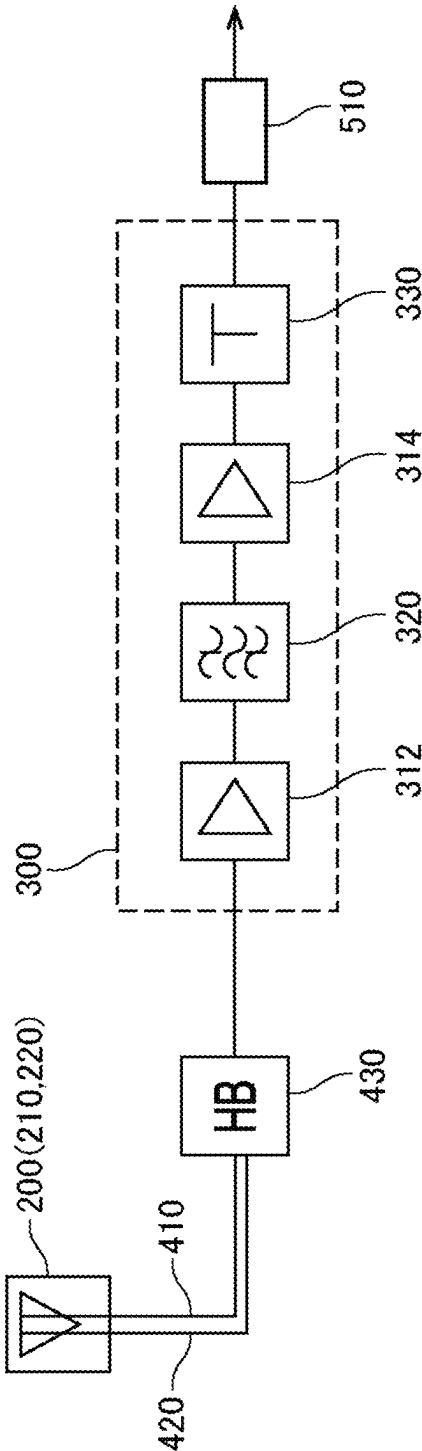


FIG. 15

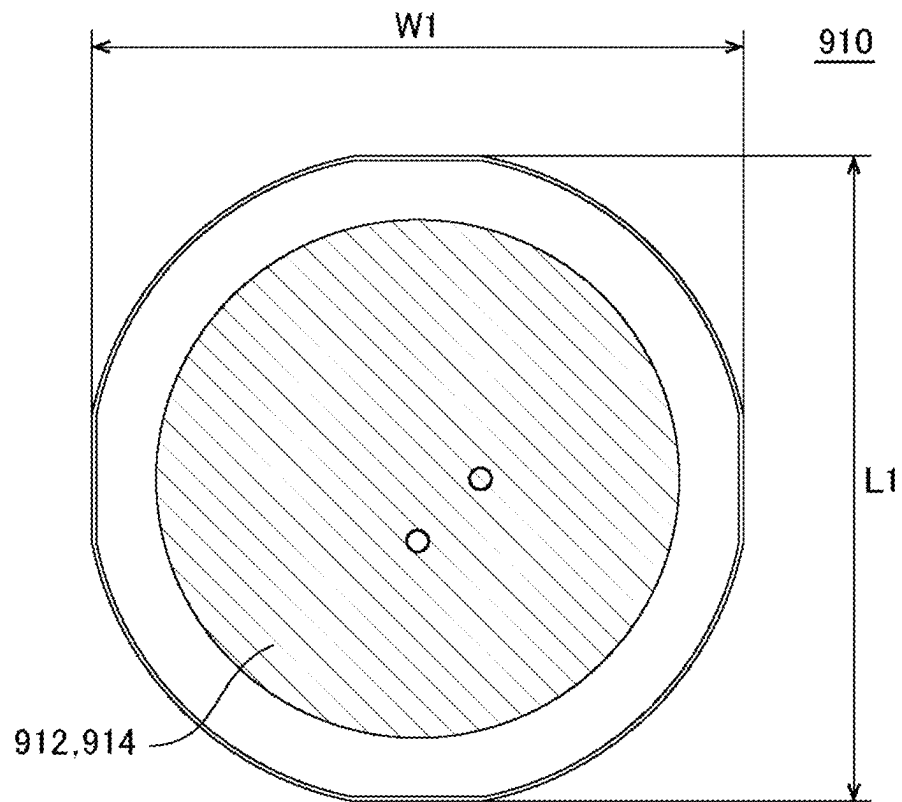


FIG. 16

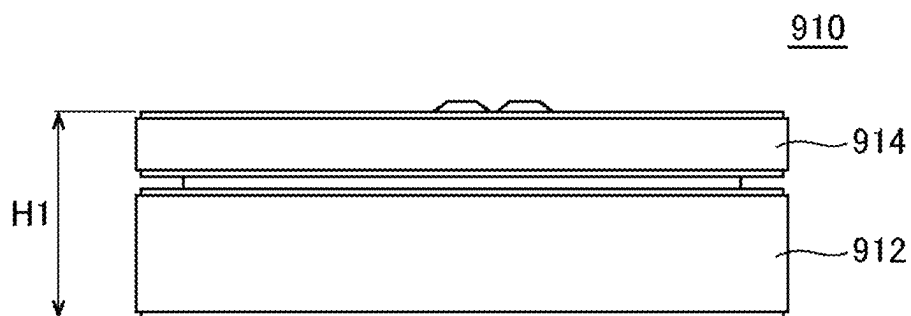


FIG. 17

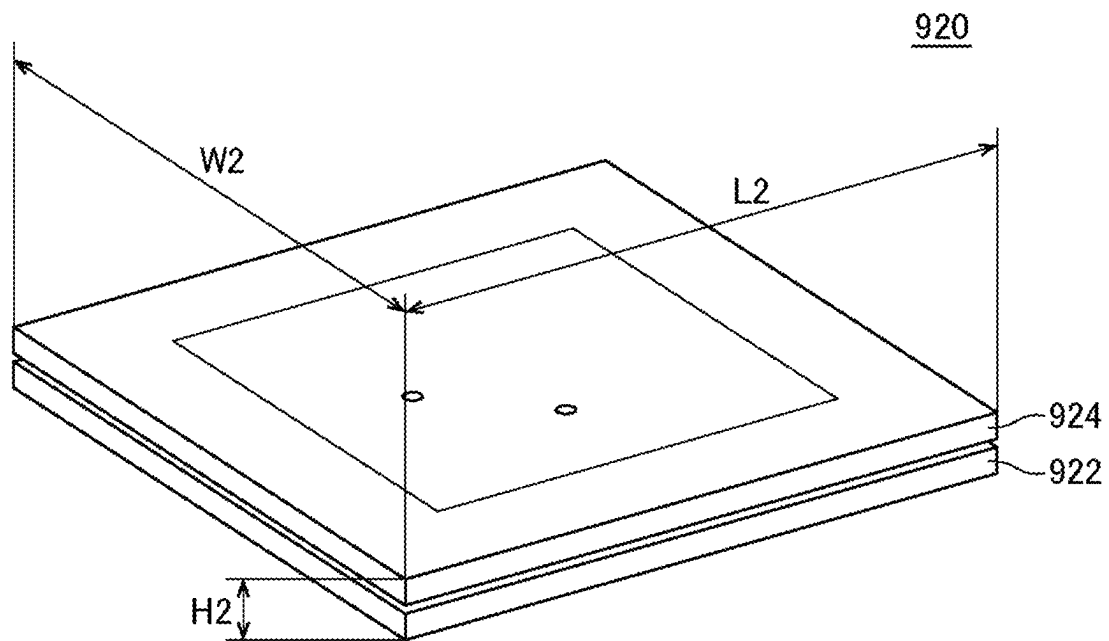


FIG. 18

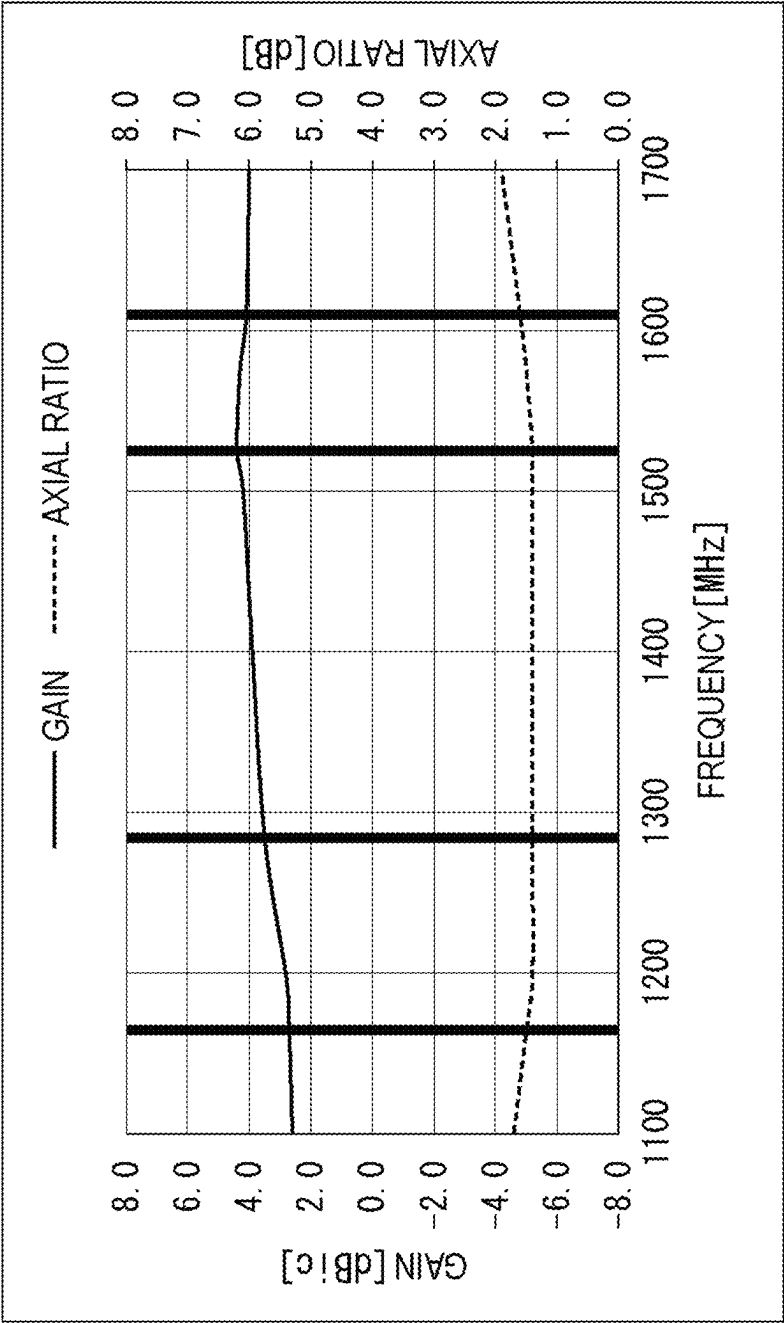


FIG. 19

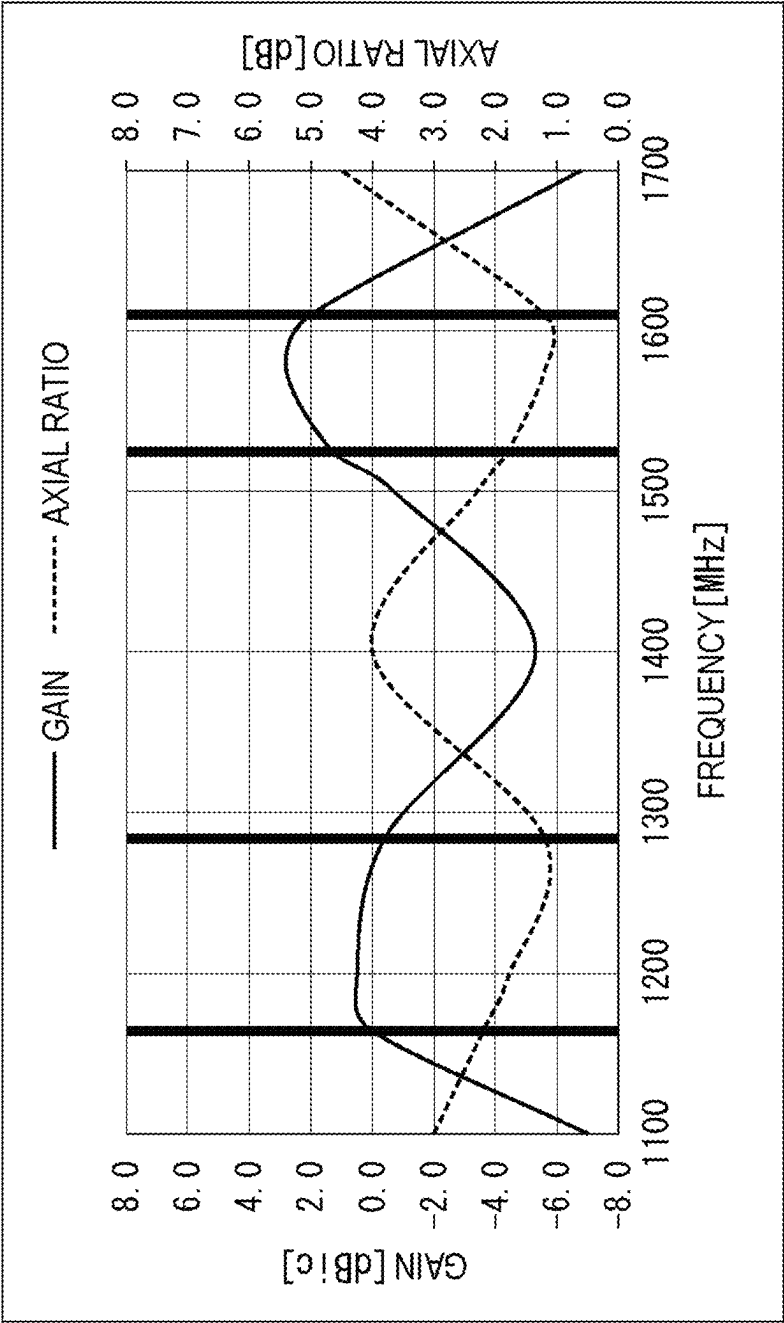
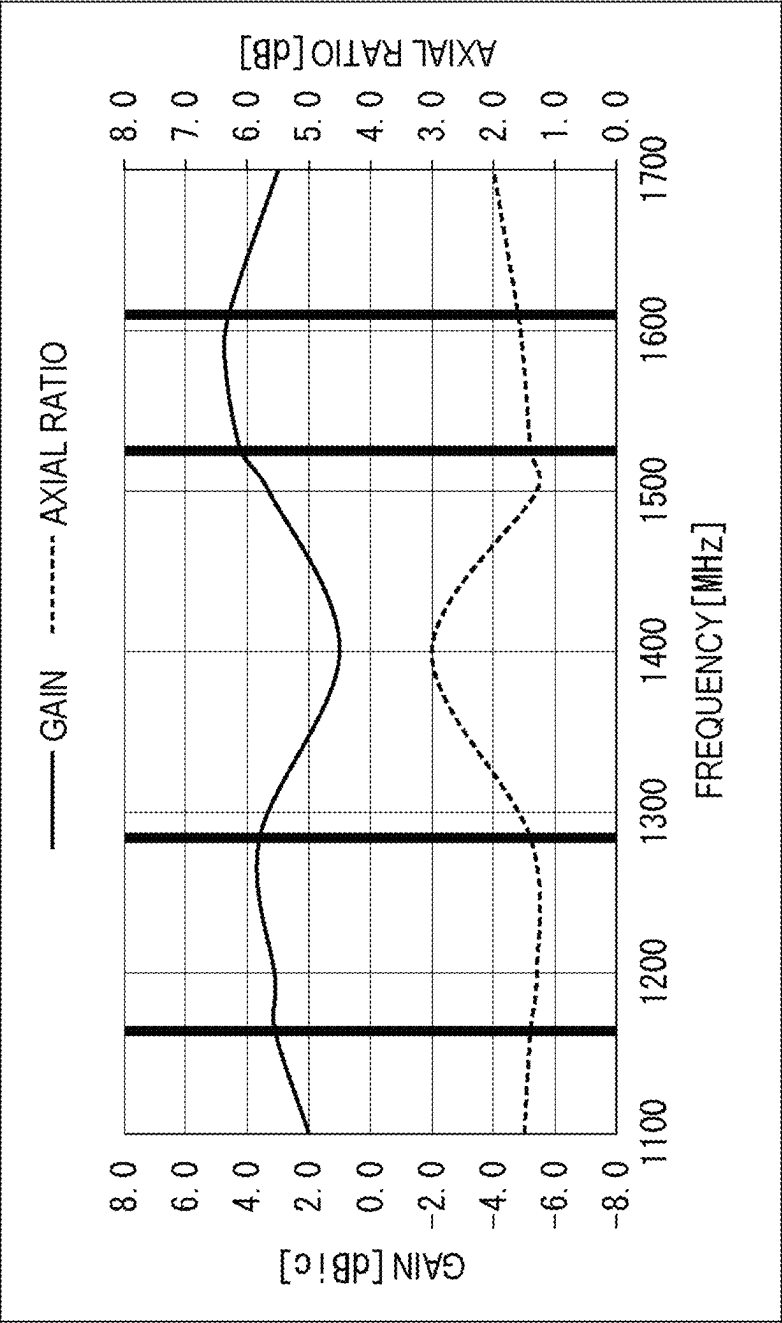


FIG. 20



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VEHICULAR ANTENNA DEVICE**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is based on PCT filing PCT/JP2021/000303, filed Jan. 7, 2021, which claims priority to JP 2020-011871, filed Jan. 28, 2020, the entire contents of each are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a vehicular antenna device.

BACKGROUND ART

In recent years, antenna devices used for Global Navigation Satellite System (GNSS) such as a Global Positioning System (GPS) have been developed. Particularly in recent years, for applications such as Advanced Driver-Assistance Systems (ADAS) that acquire high-precision position information in a short time, there is a demand for GNSS antennas that support a wide range of bands within a multi-band including L1 band (1559 MHz to 1610 MHz), L band (1525 MHz to 1559 MHz), L5 band (1164 MHz to 1214 MHz), L2 band (1212 MHz to 1254 MHz), and L6 band (1273 MHz to 1284 MHz).

Patent Documents 1 and 2 describe a stacked patch antenna. This antenna includes a first patch antenna and a second patch antenna. The first patch antenna is stacked on the second patch antenna. The center frequency of the first patch antenna is adjusted to a frequency (for example, 2.320 GHz to 2.345 GHz) used in Satellite Digital Audio Radio Service (SDARS). The center frequency of the second patch antenna is adjusted to a frequency (for example, 1.575 GHz) used in GPS.

Patent Document 3 describes a GPS patch antenna. This antenna includes a dielectric plate, a first antenna disposed on the dielectric plate and compatible with the L1 band, and a second antenna disposed on the dielectric plate and compatible with the L2 band. The size of the dielectric plate is a square plate having a length of 40 mm, a width of 40 mm, and a height of 4 mm. The relative permittivity of the dielectric plate is 6.8. The first antenna and the second antenna are formed in a loop shape. The first antenna is located inside the loop of the second antenna. Patent Document 3 describes that according to this antenna, a high gain of 0 dBic or more and a low axial ratio of 5 dB or less are obtained in the L1 band and the L2 band.

Patent Document 4 describes a multi-band GNSS patch antenna. This antenna includes a dielectric plate, a first conductive plate disposed on one surface side of the dielectric plate, and a second conductive plate disposed on the opposite surface side of one surface of the dielectric plate. A notch is formed on the outer periphery of the second conductive plate. Patent Document 4 describes that the antenna can be compatible with 1.15 GHz, 1.56 GHz, 1.17645 GHz and 1.57542 GHz by adjusting various conditions such as the shape of the notch of the second conductive plate.

RELATED DOCUMENT**Patent Document**

[Patent Document 1] U.S. Pat. No. 7,277,056
[Patent Document 2] U.S. Pat. No. 7,528,780

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[Patent Document 3] Japanese Unexamined Patent Publication No. 2018-182362

[Patent Document 4] Japanese Unexamined Patent Publication No. 2019-041240

SUMMARY OF THE INVENTION**Technical Problem**

As described above, various applications such as ADAS require a GNSS antenna compatible with a wide range of bands within a multi-band including L1 band, L band, L5 band, L2 band, and L6 band. In addition, it is desirable that such a GNSS antenna be miniaturized due to various demands such as accommodation space. When the patch antennas described in Patent Documents 1 to 4 support a wide band, however, it may be necessary to increase the size of the patch antenna and difficult to miniaturize the patch antenna. According to the patch antenna described in Patent Document 3, for example, the antenna is optimized with the L1 band and the L2 band as resonance frequencies. Accordingly, high gain and low axial ratio are obtained in the L1 band and the L2 band. In bands other than the L1 band and the L2 band, on the other hand, a high gain such as 0 dBic or more and a low axial ratio such as 5 dB or less are not obtained. Further, in this patch antenna, the antenna is miniaturized optimizing the antenna with the L1 band and the L2 band being resonance frequencies, and the compatible band range of the antenna is narrowed.

An example of an object of the present invention is to miniaturize a GNSS antenna compatible with a wide range of bands within a multi-band including the L1 band, the L band, the L5 band, the L2 band and the L6 band. Other objects of the present invention will become apparent from the description herein.

Solution to Problem

According to one aspect of the present invention, there is provided a vehicular antenna device including:

an antenna element operable in at least two or more of frequency bands including L1 band, L band, L5 band, L2 band, and L6 band and receiving circular polarization, in which

the antenna element includes

a first element having a first feed portion, and a first element section and a second element section arranged at opposite sides of the first feed portion, and a second element having a second feed portion, and a third element section and a fourth element section arranged at opposite sides of the second feed portion, at least a part of the first element and at least a part of the second element face each other, and one of the first element section and the second element section is arranged at an angle with respect to the other of the first element section and the second element section.

Advantageous Effects of Invention

According to the above aspect of the present invention, a GNSS antenna compatible with a wide range of bands within a multi-band including L1 band, L band, L5 band, L2 band, and L6 band can be miniaturized.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a vehicular antenna device according to an embodiment.

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FIG. 2 is a diagram with a cover and a ground plate removed from FIG. 1.

FIG. 3 is a diagram for explaining the details of the inclination of each of a first element section and a second element section with respect to a direction parallel to a mounting surface of a base in the example shown in FIG. 2.

FIG. 4 is a diagram showing a first modification example of FIG. 3.

FIG. 5 is a diagram showing a second modification example of FIG. 3.

FIG. 6 is a diagram showing a third modification example of FIG. 3.

FIG. 7 is a diagram for explaining the details of the region between facing portions of the first element and the second element.

FIG. 8 is a block diagram showing a first example of the details of a circuit portion shown in FIG. 2.

FIG. 9 is a block diagram showing a second example of the details of the circuit portion shown in FIG. 2.

FIG. 10 is a block diagram showing a third example of the details of the circuit portion shown in FIG. 2.

FIG. 11 is a block diagram showing a fourth example of the details of the circuit portion shown in FIG. 2.

FIG. 12 is a block diagram showing a fifth example of the details of the circuit portion shown in FIG. 2.

FIG. 13 is a block diagram showing a sixth example of the details of the circuit portion shown in FIG. 2.

FIG. 14 is a block diagram showing a seventh example of the details of a circuit portion 300 shown in FIG. 2.

FIG. 15 is a top view of a first stacked patch antenna according to comparative embodiment 1.

FIG. 16 is a side view of the first stacked patch antenna shown in FIG. 15.

FIG. 17 is a perspective view of a second stacked patch antenna according to comparative embodiment 2.

FIG. 18 is a graph showing frequency characteristics of gain and axial ratio of the antenna element (FIG. 2) according to the embodiment in the range of 1100 MHz to 1700 MHz.

FIG. 19 is a graph showing frequency characteristics of gain and axial ratio of the first stacked patch antenna (FIGS. 15 and 16) according to comparative embodiment 1 at 1100 MHz to 1700 MHz.

FIG. 20 is a graph showing frequency characteristics of gain and axial ratio of the second stacked patch antenna (FIG. 17) according to comparative embodiment 2 at 1100 MHz to 1700 MHz.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the present invention will be described with reference to the drawings. In all drawings, similar components are designated by the same reference numerals, and description thereof will not be repeated as appropriate.

In the present specification, the ordinal numbers such as “first”, “second”, and “third” are added only to distinguish the components having the same names unless otherwise specified, and does not mean a particular feature (for example, order or importance) of the component.

FIG. 1 is a perspective view of a vehicular antenna device 10 according to the embodiment. FIG. 2 is a diagram with a cover 500 and a ground plate 600 removed from FIG. 1.

In FIGS. 1 and 2, a first direction X is the front-rear direction of the vehicular antenna device 10. The positive direction (the direction indicated by the arrow indicating the first direction X) of the first direction X is the front direction

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of the vehicular antenna device 10. The negative direction (the direction opposite to the direction indicated by the arrow indicating the first direction X) of the first direction X is the rear direction of the vehicular antenna device 10. The second direction Y is the left-right direction of the vehicular antenna device 10. The second direction Y intersects the first direction X, and is specifically orthogonal to the first direction X. The positive direction (the direction indicated by the arrow indicating the second direction Y) of the second direction Y is the left direction of the vehicular antenna device 10 as viewed from the rear (the negative direction of the first direction X) of the vehicular antenna device 10. The negative direction (the direction opposite to the direction indicated by the arrow indicating the second direction Y) of the second direction Y is the right direction of the vehicular antenna device 10 as viewed from the rear (the negative direction of the first direction X) of the vehicular antenna device 10. A third direction Z is the vertical direction of the vehicular antenna device 10. The third direction Z intersects both the first direction X and the second direction Y, and is specifically orthogonal to both the first direction X and the second direction Y. The positive direction (the direction indicated by the arrow indicating the third direction Z) of the third direction Z is the upward direction of the vehicular antenna device 10. The negative direction (the direction opposite to the direction indicated by the arrow indicating the third direction Z) of the third direction Z is the downward direction of the vehicular antenna device 10. The same applies to the following figures.

The outline of the vehicular antenna device 10 will be described with reference to FIGS. 1 and 2.

As shown in FIG. 2, the vehicular antenna device 10 includes a base 100, an antenna element 200, the circuit portion 300, a first feed line 410, and a second feed line 420. As shown in FIG. 1, the vehicular antenna device 10 further includes the cover 500.

In the example shown in FIG. 1, the vehicular antenna device 10 is disposed on the ground plate 600. In the present embodiment, the ground plate 600 is the roof of an automobile. That is, the vehicular antenna device 10 is attached to the upper surface side of the roof (ground plate 600) of the automobile. When the vehicular antenna device 10 is attached to the roof of an automobile, the positive direction of the first direction X is the forward direction of the automobile, and the negative direction of the first direction X is the backward direction of the automobile. As will be described in detail later, in a case where the vehicular antenna device 10 (antenna element 200) is disposed on the ground plate 600, the antenna element 200 can operate better as a GNSS antenna as compared with the case where the vehicular antenna device 10 (antenna element 200) is not disposed on the ground plate 600. However, the ground plate 600 to which the vehicular antenna device 10 is attached is not limited to the roof of the automobile.

The base 100 has a first base member 110 and a second base member 120. The first base member 110 and the second base member 120 have a thickness in the vertical direction (third direction Z) of the vehicular antenna device 10. The first base member 110 is made of an insulating material such as a resin. The second base member 120 is located on the first base member 110. The second base member 120 is made of a conductive material such as metal. The length of each of the first base member 110, the second base member 120, and the base 100 in the front-rear direction (first direction X) of the vehicular antenna device 10 is longer than the length of each of the first base member 110, the second base

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member **120**, and the base **100** in the left-right direction (second direction Y) of the vehicular antenna device **10**.

The base **100** may be composed of only the second base member **120**, or may be composed of the second base member **120** and a metal plate. Further, the base **100** may be composed of the first base member **110** and a metal plate, or may be composed of the first base member **110**, the second base member **120** and a metal plate.

The antenna element **200** is mounted on the mounting surface **122** of the base **100** (second base member **120**) via the first feed line **410** and the second feed line **420**. Details of the first feed line **410** and the second feed line **420** will be described later. The antenna element **200** is operable in a frequency band including L1 band (1559 MHz to 1610 MHz), L band (1525 MHz to 1559 MHz), L5 band (1164 MHz to 1214 MHz), L2 band (1212 MHz to 1254 MHz), and L6 band (1273 MHz to 1284 MHz) and receives circular polarization. For example, the gain and axial ratio of the antenna element **200** in each of the L1 band, the L band, the L5 band, the L2 band, and the L6 band are 2.0 dBic or more and 4.0 dB or less, respectively. The antenna element **200** may not be operable in all of the L1 band, L band, L5 band, L2 band and L6 band, and may be operable in at least two or more of these bands.

The antenna element **200** has two elements, that is, a first element **210** and a second element **220**.

The first element **210** has two element sections, that is, a first element section **210a** and a second element section **210b**. The second element **220** has two element sections, that is, a third element section **220a** and a fourth element section **220b**.

The first element section **210a** includes two arms, that is, a first arm **212a** and a second arm **212b**. The second element section **210b** includes two arms, that is, a third arm **212c** and a fourth arm **212d**. The third element section **220a** includes two arms, that is, a fifth arm **222a** and a sixth arm **222b**. The fourth element section **220b** includes two arms, that is, the seventh arm **222c** and the eighth arm **222d**.

As will be described in detail later, the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d** include a first portion **214a**, a second portion **214b**, a third portion **214c**, and a fourth portion **214d**, respectively. As will be described in detail later, the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d** include a fifth portion **224a**, a sixth portion **224b**, a seventh portion **224c**, and an eighth portion **224d**, respectively.

Each of the first element section **210a** and the second element section **210b** of the first element **210** and the third element section **220a** and the fourth element section **220b** of the second element **220** are made of a conductive plate, specifically, a sheet metal. The first element section **210a** and the second element section **210b** of the first element **210** each have a portion operating as a self-similar antenna or an equivalent antenna. The third element section **220a** and the fourth element section **220b** of the second element **220** each have a portion operating as a self-similar antenna or an equivalent antenna. The “self-similar antenna” is an antenna such as a biconical antenna or a bow tie antenna whose shape becomes similar even if the scale (size ratio) is changed. Details of the portion operating as a self-similar antenna or an equivalent antenna will be described later.

The circuit portion **300** is mounted on the mounting surface **122** of the base **100**. The circuit portion **300** has, for example, an integrated circuit (IC). The circuit portion **300**

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is electrically connected to the first element **210** and the second element **220** via the first feed line **410** and the second feed line **420**, respectively.

The lower end of the first feed line **410** (the end on the negative direction side of the third direction Z) and the lower end of the second feed line **420** (the end on the negative direction side of the third direction Z) are physically and electrically connected to the circuit portion **300** via solder, for example. The upper end of the first feed line **410** (the end on the positive direction side of the third direction Z) and the upper end of the second feed line **420** (the end on the positive direction side of the third direction Z) are physically and electrically connected to the first element **210** and the second element **220**, for example, via solder, respectively. Each of the first feed line **410** and the second feed line **420** is, for example, a coaxial line. The first feed line **410** extends parallel to the height direction of the vehicular antenna device **10** between the lower end of the first feed line **410** and the upper end of the first feed line **410**. The second feed line **420** extends parallel to the height direction of the vehicular antenna device **10** between the lower end of the second feed line **420** and the upper end of the second feed line **420**.

In the present embodiment, the first feed line **410** and the second feed line **420** are supports for supporting the first element **210** and the second element **220** on the mounting surface **122** of the base **100**. However, the method of supporting the first element **210** and the second element **220** on the mounting surface **122** of the base **100** is not limited to this example.

The length of each of the first feed line **410** and the second feed line **420** in the height direction of the vehicular antenna device **10** is approximately $\lambda/4$ (λ : wavelength at the operating frequency of the antenna element **200**). Approximately $\lambda/4$ means not only exact $\lambda/4$ but also a length slightly deviated from $\lambda/4$ (for example, a length within the range of $\pm\lambda/10$ from $\lambda/4$). The operating frequency of the antenna element **200** is, for example, the central frequency in each of the L1 band, the L band, the L5 band, the L2 band, and the L6 band. However, the operating frequency of the antenna element **200** may not be the central frequency in these bands, and may be a frequency deviating from the central frequency. The length of each of the first feed line **410** and the second feed line **420** in the height direction of the vehicular antenna device **10** is approximately $\lambda/4$. Further, in a case where the vehicular antenna device **10** is disposed on the ground plate **600**, the antenna element **200** can operate better as a GNSS antenna as compared with the case where the vehicular antenna device **10** is not disposed on the ground plate **600**. However, the vehicular antenna device **10** may not be disposed on the ground plate **600**.

The cover **500** is attached to the upper surface side of the first base member **110** of the base **100**, and covers the second base member **120** of the base **100**, the antenna element **200**, the circuit portion **300**, the first feed line **410**, and the second feed line **420**. For attaching the cover **500** and the base **100**, fixing means such as bolts may be adopted, or fixing means such as welding or adhesion may be adopted. As will be described in detail later, in the present embodiment, as compared with the case where the first element section **210a** and the second element section **210b** of the antenna element **200** are disposed in parallel with the mounting surface **122** of the base **100**, the length of the antenna element **200** in the left-right direction of the vehicular antenna device **10** is shortened. Accordingly, the length of the cover **500** in the left-right direction of the vehicular antenna device **10** can be shortened as compared with the case where the first element

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section **210a** and the second element section **210b** are disposed in parallel with the mounting surface **122** of the base **100**.

Next, the details of the shape of the first element **210** will be described with reference to FIG. 2.

The first element section **210a** and the second element section **210b** face each other in a predetermined direction, specifically, in the left-right direction of the vehicular antenna device **10**. Specifically, the first element section **210a** is located on the left side of the second element section **210b**, and the second element section **210b** is located on the right side of the first element section **210a**. However, the facing direction of the first element section **210a** and the second element section **210b** is not limited to the left-right direction of the vehicular antenna device **10**, and may be, for example, the front-rear direction of the vehicular antenna device **10**. The first element **210** is located above the second element **220** in the height direction of the vehicular antenna device **10**. However, the first element **210** may be located below the second element **220** in the height direction of the vehicular antenna device **10**.

The first element **210** has a first feed portion that is a portion connected to the upper end of the first feed line **410**. Each of the first element section **210a** and the second element section **210b** has a first base end portion where the first element section **210a** and the second element section **210b** are closest to each other and which includes the first feed portion. A first central portion, which is a substantially intermediate point between the first base end portion of the first element section **210a** and the first base end portion of the second element section **210b**, is the first feed portion of the first element **210**. The “substantially intermediate point” means not only the exact intermediate point between the first base end portion of the first element section **210a** and the first base end portion of the second element section **210b**, but also a point deviated from the exact intermediate point by a slight distance (for example, 5% or less of the distance between the first base end portion of the first element section **210a** and the first base end portion of the second element section **210b**).

The first arm **212a** and the second arm **212b** of the first element section **210a** are arranged in the front-rear direction of the vehicular antenna device **10**. Specifically, the first arm **212a** is located on the front side of the second arm **212b**, and the second arm **212b** is located on the rear side of the first arm **212a**. The first arm **212a** and the second arm **212b** extend in a direction away from each other from the first base end portion of the first element section **210a**. The third arm **212c** and the fourth arm **212d** of the second element section **210b** are arranged in the front-rear direction of the vehicular antenna device **10**. Specifically, the third arm **212c** is located on the front side of the fourth arm **212d**, and the fourth arm **212d** is located on the rear side of the third arm **212c**. The third arm **212c** and the fourth arm **212d** extend in a direction away from each other from the first base end portion of the second element section **210b**. Further, the first arm **212a** and the third arm **212c** are arranged in the left-right direction of the vehicular antenna device **10**. The second arm **212b** and the fourth arm **212d** are arranged in the left-right direction of the vehicular antenna device **10**. The first element **210** is formed by joining the first element section **210a** and the second element section **210b** symmetrically about the first feed portion of the first element **210**. Alternatively, the first element section **210a** and the second element section **210b** may be integrally formed.

The tips of the first arm **212a** and the second arm **212b** of the first element section **210a** (opposite ends of the first base

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end portion) are spaced apart from each other and are open. That is, the tips of the first arm **212a** and the second arm **212b** of the first element section **210a** each have an open end portion. Each open end portion of the first element section **210a** is formed to mainly secure the area of the first element **210** above a certain level in order to secure a low frequency band, particularly to enable use in a lower frequency band. In the present embodiment, each open end portion of the first element section **210a** has an L-shape. However, the shape of each open end portion of the first element section **210a** is not limited to the L-shape, and may be, for example, a trapezoid, a rhombus, an ellipse, a circle, a triangle, or the like. A shape similar to the shape described above for the tip of each of the first arm **212a** and the second arm **212b** of the first element section **210a** may be applied to the tip of each of the third arm **212c** and the fourth arm **212d** of the second element section **210b**.

The width of each of the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d** increases from the first base end portion toward the tip of each of the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d**. Accordingly, the width of each of the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d** in the region far from the first base end portion is wider than the width of each of the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d** in the region close to the first base end portion. The distance between the first arm **212a** and the third arm **212c** increases continuously or stepwise from the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b** toward the tip of the first arm **212a** or the tip of the third arm **212c**. Accordingly, the distance between the first arm **212a** and the third arm **212c** in the region far from the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b** is wider than the distance between the first arm **212a** and the third arm **212c** in the region close to the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b**. Similarly, the distance between the second arm **212b** and the fourth arm **212d** increases continuously or stepwise from the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b** toward the tip of the second arm **212b** or the tip of the fourth arm **212d**. Accordingly, the distance between the second arm **212b** and the fourth arm **212d** in the region far from the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b** is wider than the distance between the second arm **212b** and the fourth arm **212d** in the region close to the first base end portion of the first element section **210a** or the first base end portion of the second element section **210b**. In this way, the first arm **212a**, the second arm **212b**, the third arm **212c**, and the fourth arm **212d** of the first element **210** operate as a self-similar antenna or an equivalent antenna.

In the present embodiment, the first element section **210a** is formed in a substantially C-shape. The substantially C-shape is a shape formed by subtracting a portion from a substantially circular shape such as a circle or an ellipse. The first element section **210a** has an opening formed between the first arm **212a** and the second arm **212b** by this substantially C-shape. The first element section **210a** may be formed in a substantially U-shape, a substantially V-shape, or a substantially n-shape. The substantially U-shape is, for example, a shape formed by subtracting a portion from a substantially quadrangle and rounding the opposite side of

the portion of the substantially quadrangle. The substantially V-shape is, for example, a shape formed by subtracting a portion from a substantially triangle or a portion from a substantially quadrangle such as a trapezoid whose upper side is relatively short with respect to the lower side. The substantially n-shape is, for example, a shape formed by subtracting a portion from a substantially quadrangle such as a rectangle, a square, or a trapezoid whose upper side is relatively long with respect to the lower side. The same applies to the shape of the second element section **210b**. In the first element section **210a** and the second element section **210b**, the opening of the first element section **210a** and the opening of the second element section **210b** are disposed to face opposite to each other.

As will be described later with reference to FIG. 3, the first element section **210a** and the second element section **210b** are tilted by substantially the same angle from the direction (second direction Y) parallel to the mounting surface **122** of the base **100** toward the side (the negative direction side of the third direction Z) where the mounting surface **122** of the base **100** is located. In this way, one of the first element section **210a** and the second element section **210b** is arranged at an angle with respect to the other of the first element section **210a** and the second element section **210b**. Substantially the same angle means that the angle of inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** and the angle of inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** not only are exactly the same but also differ by a slight angle (for example, +5 degrees or less).

According to the present embodiment, the length of the first element **210** in the facing direction of the first element section **210a** and the second element section **210b** can be shortened, as compared to the case where one of the first element section **210a** and the second element section **210b** is not tilted with respect to the other of the first element section **210a** and the second element section **210b**, such as the case where the first element section **210a** and the second element section **210b** are disposed in parallel with the mounting surface **122** of the base **100**. Accordingly, the vehicular antenna device **10** can be miniaturized. According to the present embodiment, the length of the first element **210** in the height direction of the vehicular antenna device **10** can be shortened, as compared with the case where the first element section **210a** and the second element section **210b** are tilted toward the opposite side (the positive direction side of the third direction Z) to the side where the mounting surface **122** is located. Further, according to the present embodiment, the radiation directivity of the antenna element **200** in the zenith direction (positive direction of the third direction Z) can be strengthened, as compared with the case where the angle of inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** is different from the angle of inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122**.

In the present embodiment, the first element section **210a** is disposed on a plane inclined in the positive direction of the second direction Y with respect to the third direction Z. However, the first element section **210a** may not be disposed on this plane. For example, the first element section **210a** may be at least partially curved or bent when viewed from the positive or negative direction of the first direction X. The same applies to the second element section **210b**.

Next, the details of the shape of the second element **220** will be described with reference to FIG. 2. In the present

embodiment, the size connecting the outer edges of the second element **220** (hereinafter referred to as “outer edge size”) is substantially equal to the outer edge size of the first element **210**. That is, the first element **210** and the second element **220** have substantially the same shape. “The outer edge sizes are substantially equal” means that the outer edge size of the second element **220** is not only exactly equal to the outer edge size of the first element **210**, but also, for example, 95% or more and 105% or less of the outer edge size of the first element **210**. However, the outer edge size of the second element **220** may be different from the outer edge size of the first element **210**.

The fifth portion **224a**, the sixth portion **224b**, the seventh portion **224c**, and the eighth portion **224d** are turned by about 90 degrees with respect to the third portion **214c**, the fourth portion **214d**, the first portion **214a**, and the second portion **214b**, respectively. “About 90 degrees” means that the turning angle of the first element **210** with respect to the second element **220** is not only exactly 90 degrees, but also deviates from 90 degrees by a slight angle (for example, +2.5 degrees or less).

The third element section **220a** and the fourth element section **220b** are arranged in the front-rear direction of the vehicular antenna device **10**. Specifically, the third element section **220a** is located on the front side of the fourth element section **220b**, and the fourth element section **220b** is located on the rear side of the third element section **220a**.

The second element **220** has a second feed portion that is a part connected to the upper end of the second feed line **420**. Each of the third element section **220a** and the fourth element section **220b** has a second base end portion in which the third element section **220a** and the fourth element section **220b** are closest to each other and which includes the second feed portion. A second central portion, which is a substantially intermediate point between the second base end portion of the third element section **220a** and the second base end portion of the fourth element section **220b**, is the second feed portion of the second element **220**. The “substantially intermediate point” means not only the exact intermediate point between the second base end portion of the third element section **220a** and the second base end portion of the fourth element section **220b**, but also a point deviated from the exact intermediate point by a slight distance (for example, 5% or less of the distance between the second base end portion of the third element section **220a** and the second base end portion of the fourth element section **220b**).

The fifth arm **222a** and the sixth arm **222b** of the third element section **220a** are arranged in the left-right direction of the vehicular antenna device **10**. Specifically, the fifth arm **222a** is located on the left side of the sixth arm **222b**, and the sixth arm **222b** is located on the right side of the fifth arm **222a**. The fifth arm **222a** and the sixth arm **222b** extend in a direction away from each other from the second base end portion of the third element section **220a**. The seventh arm **222c** and the eighth arm **222d** of the fourth element section **220b** are arranged in the left-right direction of the vehicular antenna device **10**. Specifically, the seventh arm **222c** is located on the left side of the eighth arm **222d**, and the eighth arm **222d** is located on the right side of the seventh arm **222c**. The seventh arm **222c** and the eighth arm **222d** extend in a direction away from each other from the second base end portion of the fourth element section **220b**. Further, the fifth arm **222a** and the seventh arm **222c** are arranged in the front-rear direction of the vehicular antenna device **10**. The sixth arm **222b** and the eighth arm **222d** are arranged in the front-rear direction of the vehicular antenna device **10**. The

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second element **220** is formed by joining the third element section **220a** and the fourth element section **220b** symmetrically about the second feed portion of the second element **220**. Alternatively, the third element section **220a** and the fourth element section **220b** may be integrally formed.

The tips of the fifth arm **222a** and the sixth arm **222b** of the third element section **220a** (opposite ends of the second base end portion) are spaced apart from each other and are open. That is, the tips of the fifth arm **222a** and the sixth arm **222b** of the third element section **220a** each have an open end portion. Each open end portion of the third element section **220a** is formed to mainly secure the area of the second element **220** above a certain level in order to secure a low frequency band, particularly to enable use in a lower frequency band. In the present embodiment, each open end portion of the third element section **220a** has an L-shape. However, the shape of each open end portion of the third element section **220a** is not limited to the L-shape, and may be, for example, a trapezoid, a rhombus, an ellipse, a circle, a triangle, or the like. A shape similar to the shape described above for the tip of each of the fifth arm **222a** and the sixth arm **222b** of the third element section **220a** may be applied to the tip of each of the seventh arm **222c** and the eighth arm **222d** of the fourth element section **220b**.

The width of each of the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d** increases from the second base end portion toward the tip of each of the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d**. Accordingly, the width of each of the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d** in the region far from the second base end portion is wider than the width of each of the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d** in the region close to the second base end portion. The distance between the fifth arm **222a** and the seventh arm **222c** increases continuously or stepwise from the second base end portion of the third element section **220a** or the second base end portion of the fourth element section **220b** toward the tip of the fifth arm **222a** and the tip of the seventh arm **222c**. Accordingly, the distance between the fifth arm **222a** and the seventh arm **222c** in the region far from the second base end portion of the third element section **220a** or the second base end portion of the fourth element section **220b** is wider than the distance between the fifth arm **222a** and the seventh arm **222c** in the region close to the second base end portion of the third element section **220a** or the second base end portion of the fourth element section **220b**. Similarly, the distance between the sixth arm **222b** and the eighth arm **222d** increases continuously or stepwise from the second base end portion of the second element section **210b** or the second base end portion of the fourth element section **220b** toward the tip of the sixth arm **222b** or the tip of the eighth arm **222d**. Accordingly, the distance between the sixth arm **222b** and the eighth arm **222d** in the region far from the second base end portion of the third element section **220a** or the second base end portion of the fourth element section **220b** is wider than the distance between the sixth arm **222b** and the eighth arm **222d** in the region close to the second base end portion of the third element section **220a** or the first base end portion of the fourth element section **220b**. In this way, the fifth arm **222a**, the sixth arm **222b**, the seventh arm **222c**, and the eighth arm **222d** of the second element **220** operate as a self-similar antenna or an equivalent antenna.

When each of the first element **210** and the second element **220** includes a part operating as a self-similar antenna or an equivalent antenna, the antenna element **200**

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operates as, for example, a tapered slot antenna in a relatively high frequency band, and operates as, for example, a loop antenna in a relatively low frequency band. In a specific frequency band in an intermediate frequency band between a relatively high frequency band and a relatively low frequency band, the antenna element **200** operates as a dipole antenna. In each band between the relatively high frequency band, the relatively low frequency band, and the intermediate frequency band, the antenna element **200** operates with the operating principles of these antennas combined, that is, an integrated antenna. Accordingly, the antenna element **200** can operate stably over a wide frequency band even though it is one antenna element.

In the present embodiment, each of the third element section **220a** and the fourth element section **220b** is formed, for example, in one of a substantially C-shape, a substantially U-shape, a substantially V-shape, or a substantially n-shape, as described above for the first element section **210a**.

The rate of change (rate of increase) of the width of each arm of the second element **220** from the second base end portion to the tip of each arm may be different from the rate of change (rate of increase) of the width of each arm of the first element **210** from the first base end portion to the tip of each arm. For example, the rate of change (rate of increase) of the width of each arm of the second element **220** may be smaller than the rate of change (rate of increase) of the width of each arm of the first element **210**.

The first arm **212a** and the fifth arm **222a** include facing parts, that is, the first portion **214a** and the fifth portion **224a**, respectively. The first portion **214a** of the first element section **210a** and the fifth portion **224a** of the third element section **220a** face each other. Specifically, the first portion **214a** of the first element section **210a** and the fifth portion **224a** of the third element section **220a** face each other substantially in parallel.

The second arm **212b** and the seventh arm **222c** include facing parts, that is, the second portion **214b** and the seventh portion **224c**, respectively. The second portion **214b** of the first element section **210a** and the seventh portion **224c** of the fourth element section **220b** face each other. Specifically, the second portion **214b** of the first element section **210a** and the seventh portion **224c** of the fourth element section **220b** face each other substantially in parallel.

The third arm **212c** and the sixth arm **222b** include facing parts, that is, the third portion **214c** and the sixth portion **224b**, respectively. The third portion **214c** of the first element section **210a** and the sixth portion **224b** of the third element section **220a** face each other. Specifically, the third portion **214c** of the first element section **210a** and the sixth portion **224b** of the third element section **220a** face each other substantially in parallel.

The fourth arm **212d** and the eighth arm **222d** include facing parts, that is, the fourth portion **214d** and the eighth portion **224d**, respectively. The fourth portion **214d** of the first element section **210a** and the eighth portion **224d** of the fourth element section **220b** face each other. Specifically, the fourth portion **214d** of the first element section **210a** and the eighth portion **224d** of the fourth element section **220b** face each other substantially in parallel.

That each part of the first element **210** and each part of the second element **220** face each other substantially in parallel means not only that each part of the first element **210** and each part of the second element **220** face each other strictly in parallel but also that one of each part of the first element **210** and each part of the second element **220** is tilted by a slight angle (for example, +2.5 degrees or less) from the

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direction parallel to the other of each part of the first element **210** and each part of the second element **220**.

As described above, at least a part of the first element **210** and at least a part of the second element **220** face each other. More specifically, a split ring (a shape in which a part of the ring is cut out and faced) is formed in each of between the facing parts of the first element **210** and the second element **220**, that is, between the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220**, between the second portion **214b** of the first element **210** and the seventh portion **224c** of the second element **220**, between the third portion **214c** of the first element **210** and the sixth portion **224b** of the second element **220**, and between the fourth portion **214d** of the first element **210** and the eighth portion **224d** of the second element **220**. Accordingly, the compatible band of the antenna element **200** can be expanded to a relatively low frequency band side.

As described above, the first element section **210a** is turned by about 90 degrees with respect to the second element section **210b**. In this case, the direction of polarization of the first element **210** and the direction of polarization of the second element **220** are orthogonal to each other. Specifically, since the first element **210** and the second element **220** have substantially the same shape, there is almost no difference in amplitude and phase between the linear polarization of the first element **210** and the linear polarization of the second element **220**, which are orthogonal to each other, and the antenna element **200** receives circular polarization.

Next, the inclination of each of the first element section **210a** and the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100** will be described with reference to FIGS. 3 to 6. As will be described later with reference to FIGS. 3 to 6, at least one of the first element section **210a** and the second element section **210b** is tilted from the direction parallel to the mounting surface **122** of the base **100** toward the side (the negative direction side of the third direction **Z**) where the mounting surface **122** is located or the opposite side (the positive direction side of the third direction **Z**) of the side where the mounting surface **122** is located. Thereby, one of the first element section **210a** and the second element section **210b** can be arranged at an angle with respect to the other of the first element section **210a** and the second element section **210b**.

FIG. 3 is a diagram for explaining the details of the inclination of each of the first element section **210a** and the second element section **210b** with respect to a direction parallel to the mounting surface **122** of the base **100** in the example shown in FIG. 2.

In FIG. 3, a part of the base **100**, the first element **210** and the second element **220** of the antenna element **200**, and the first feed line **410** and the second feed line **420** when viewed from the front of the vehicular antenna device **10** (FIG. 1 or 2) are schematically shown. In FIG. 3, the broken line passing through the upper ends of the first feed line **410** and the second feed line **420** indicates a direction parallel to the mounting surface **122** of the base **100**. $\theta 1$ in FIG. 3 indicates the angle of inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100**. $\theta 2$ in FIG. 3 indicates the angle of inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100**. FIG. 3 does not show the second element **220**. As described above, the facing parts of the first element **210** and the second element **220** are disposed substantially in parallel

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with each other. The matters described here with respect to FIG. 3 are the same in FIGS. 4 to 6 described later.

As described above, in the example shown in FIG. 3 (FIG. 2), the angle $\theta 1$ of the inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100** and the angle $\theta 2$ of the inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100** are substantially equal to each other. Each of the angle $\theta 1$ and the angle $\theta 2$ is, for example, greater than 0 degrees and 70 degrees or less. In FIGS. 4 to 6 described later, each of the angle $\theta 1$ and the angle $\theta 2$ is, for example, greater than 0 degrees and 70 degrees or less, unless otherwise specified. When each of the angle $\theta 1$ and the angle $\theta 2$ is in the above range, the decrease from the characteristics (for example, gain or axial ratio) of the antenna element **200** with the first element section **210a** and the second element section **210b** disposed parallel to each other can be suppressed within a sufficiently acceptable range, and the length of the first element **210** in the facing direction of the first element section **210a** and the second element section **210b** can be shortened.

FIG. 4 is a diagram showing a first modification example of FIG. 3.

The first element section **210a** and the second element section **210b** are tilted by different angles from the direction parallel to the mounting surface **122** of the base **100** toward the side where the mounting surface **122** of the base **100** is located. In the example shown in FIG. 4, the angle $\theta 1$ of the inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100** is larger than the angle $\theta 2$ of the inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100**.

The angle $\theta 1$ of the inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100** may be smaller than the angle $\theta 2$ of the inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100**.

Even in the present modification example, the length of the first element **210** in the facing direction of the first element section **210a** and the second element section **210b** can be shortened, as compared to the case where one of the first element section **210a** and the second element section **210b** is not tilted with respect to the other of the first element section **210a** and the second element section **210b**, such as the case where the first element section **210a** and the second element section **210b** are disposed in parallel with the mounting surface **122** of the base **100**. Even in the present modification example, as compared with the case where the first element section **210a** and the second element section **210b** are tilted toward the opposite side to the side where the mounting surface **122** is located, the length of the first element **210** in the height direction of the vehicular antenna device **10** (FIG. 1 or FIG. 2) can be shortened. Further, in the present modification example, the radiation directivity of the antenna element **200** can be tilted from the zenith direction toward a desired direction, by adjusting the angle of inclination of each of the first element section **210a** and the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100**.

FIG. 5 is a diagram showing a second modification example of FIG. 3.

The first element section **210a** and the second element section **210b** are tilted from the direction parallel to the mounting surface **122** toward the opposite side to the side

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where the mounting surface **122** is located. In the example shown in FIG. **5**, the angle $\theta 1$ of the inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100** and the angle $\theta 2$ of the inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100** are substantially equal to each other. However, the angle $\theta 1$ of the inclination of the first element section **210a** with respect to the direction parallel to the mounting surface **122** of the base **100** and the angle $\theta 2$ of the inclination of the second element section **210b** with respect to the direction parallel to the mounting surface **122** of the base **100** may be different from each other.

Even in the present modification example, the length of the antenna element **200** in the left-right direction of the vehicular antenna device **10** (FIG. **1** or **2**) can be shortened, as compared with the case where both the first element section **210a** and the second element section **210b** are disposed in parallel with the mounting surface **122** of the base **100**.

FIG. **6** is a diagram showing a third modification example of FIG. **3**.

The first element section **210a** is disposed in parallel with the mounting surface **122** of the base **100**, while the second element section **210b** is tilted from a direction parallel to the mounting surface **122** of the base **100** toward the side where the mounting surface **122** of the base **100** is located. The first element section **210a** may be tilted from a direction parallel to the mounting surface **122** of the base **100** toward the opposite side to the side where the mounting surface **122** of the base **100** is located.

Even in the present modification example, the length of the antenna element **200** in the left-right direction of the vehicular antenna device **10** (FIG. **1** or **2**) can be shortened, as compared with the case where both the first element section **210a** and the second element section **210b** are disposed in parallel with the mounting surface **122** of the base **100**.

FIG. **7** is a diagram for explaining the details of the region between facing parts of the first element **210** and the second element **220**. FIG. **7** shows a cross-sectional view of the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220** in a direction perpendicular to the front-rear direction of the vehicular antenna device **10**.

The vehicular antenna device **10** includes an insulator **230**. The insulator **230** is disposed between the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220**. The insulator **230** is, for example, a resin. The insulator **230** can suppress contact between the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220**. The insulator **230** also serves as a spacer for adjusting the width of the region (gap) between the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220**.

The insulator **230** may be disposed not only between the first portion **214a** of the first element **210** and the fifth portion **224a** of the second element **220**, but also between the other facing parts of the first element **210** and the second element **220**, that is, between the second portion **214b** of the first element **210** and the seventh portion **224c** of the second element **220**, between the third portion **214c** of the first element **210** and the sixth portion **224b** of the second element **220**, and between the fourth portion **214d** of the first element **210** and the eighth portion **224d** of the second element **220**. The insulator **230** may not be disposed in all of these four regions, and may be disposed in at least one place of these four regions.

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FIG. **8** is a block diagram showing a first example of the details of the circuit portion **300** shown in FIG. **2**.

The circuit portion **300** includes a first-stage amplifier **312**, a first bandpass filter (first BPF) **322**, a second bandpass filter (second BPF) **324**, two second-stage amplifiers **314** (first second-stage amplifier **314a** and a second second-stage amplifier **314b**), and two attenuators **330** (first attenuator **330a** and second attenuator **330b**).

The first feed line **410** and the second feed line **420** are electrically connected to the circuit portion **300** via a hybrid circuit **430**. The hybrid circuit **430** imparts a 90 degrees phase difference to the signal sent to the first feed line **410** connected to the first element **210** and the signal sent to the second feed line **420** connected to the second element **220**. The hybrid circuit **430** combines the signal from the first feed line **410** and the signal from the second feed line **420**. The hybrid circuit **430** is provided, for example, on the lower surface (the surface on the negative direction side of the third direction Z) of the circuit portion **300**. In this example, the lower end (the end on the negative direction side of the third direction Z) of each of the first feed line **410** and the second feed line **420** is connected to the hybrid circuit **430**. In the present embodiment, the hybrid circuit **430** is present inside the circuit portion **300**. However, the hybrid circuit **430** may be present in a region different from the inside of the circuit portion **300**. The hybrid circuit **430** includes, for example, a low-pass filter portion imparting a -45 degrees phase difference to a signal and having a predetermined characteristic impedance (for example, 50 Ω), and a high-pass filter portion imparting a $+45$ degrees phase difference to a signal and having a predetermined characteristic impedance (for example, 50 Ω). These low-pass filter portion and high-pass filter portion impart a 90 degrees phase difference to the signal to the first feed line **410** and the signal to the second feed line **420**. The hybrid circuit **430** allows the antenna element **200** to receive circular polarization.

The circuit portion **300** functions as a Low Noise Amplifier (LNA). Specifically, first, the signals sent from the first feed line **410** and the second feed line **420** and combined by the hybrid circuit **430** are amplified by the first-stage amplifier **312**. The signal amplified by the first-stage amplifier **312** is sent to the first BPF **322** and the second BPF **324**. The first BPF **322** passes, for example, L1 band and L band signals. The second BPF **324** passes, for example, L5 band, L2 band and L6 band signals. The signal extracted by the first BPF **322** is amplified by the first second-stage amplifier **314a** and then sent to the first cable **510a** via the first attenuator **330a**. On the other hand, the signal passing through the second BPF **324** is amplified by the second second-stage amplifier **314b** and then sent to the second cable **510b** via the second attenuator **330b**.

FIG. **9** is a block diagram showing a second example of the details of the circuit portion **300** shown in FIG. **2**. The example shown in FIG. **9** is the same as the example shown in FIG. **8** except for the following points. That is, in the example shown in FIG. **9**, the signals sent from the first feed line **410** and the second feed line **420** and combined by the hybrid circuit **430** are sent to the parallel-connected first BPF **322** and second BPF **324**. The signal passing through the first BPF **322** and the signal passing through the second BPF **324** are amplified by the first-stage amplifier **312**, further amplified by the second-stage amplifier **314**, and sent to the cable **510** via the attenuator **330**.

FIG. **10** is a block diagram showing a third example of the details of the circuit portion **300** shown in FIG. **2**. The example shown in FIG. **10** is the same as the example shown in FIG. **9** except that the parallel-connected first BPF **322**

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and second BPF 324 are disposed between the first-stage amplifier 312 and the second-stage amplifier 314. That is, in the example shown in FIG. 10, the signals sent from the first feed line 410 and the second feed line 420 and combined by the hybrid circuit 430 are amplified by the first-stage amplifier 312 and sent to the first BPF 322 and the second BPF 324. The signal passing through the first BPF 322 and the signal passing through the second BPF 324 are amplified by the second-stage amplifier 314, and sent to the cable 510 via the attenuator 330.

FIG. 11 is a block diagram showing a fourth example of the details of the circuit portion 300 shown in FIG. 2. The example shown in FIG. 11 is the same as the example shown in FIG. 9 except that the first-stage amplifier 312 and the second-stage amplifier 314 are disposed between the hybrid circuit 430 and the parallel-connected first BPF 322 and second BPF 324. That is, in the example shown in FIG. 11, the signals sent from the first feed line 410 and the second feed line 420 and combined by the hybrid circuit 430 are amplified by the first-stage amplifier 312, are further amplified by the second-stage amplifier 314, and sent to the first BPF 322 and the second BPF 324. The signal passing through the first BPF 322 and the signal passing through the second BPF 324 are sent to the cable 510 via the attenuator 330.

FIG. 12 is a block diagram showing a fifth example of the details of the circuit portion 300 shown in FIG. 2. The example shown in FIG. 12 is the same as the example shown in FIG. 9 except that there is only a one-stage amplifier (amplifier 310), instead of the two-stage amplifiers (the first-stage amplifier 312 and the second-stage amplifier 314 in FIG. 9), and the attenuator 330 (FIG. 9) is not provided. That is, in the example shown in FIG. 12, the signals sent from the first feed line 410 and the second feed line 420 and combined by the hybrid circuit 430 are sent to the first BPF 322 and the second BPF 324. The signal passing through the first BPF 322 and the signal passing through the second BPF 324 are amplified by the amplifier 310, and sent to the cable 510.

FIG. 13 is a block diagram showing a sixth example of the details of the circuit portion 300 shown in FIG. 2. The example shown in FIG. 13 is the same as the example shown in FIG. 12 except that the amplifier 310 is disposed between the hybrid circuit 430 and the parallel-connected first BPF 322 and second BPF 324. That is, in the example shown in FIG. 13, the signals sent from the first feed line 410 and the second feed line 420 and combined by the hybrid circuit 430 are amplified by the amplifier 310 and sent to the first BPF 322 and the second BPF 324. The signal passing through the first BPF 322 and the signal passing through the second BPF 324 are sent to the cable 510 as they are.

FIG. 14 is a block diagram showing a seventh example of the details of the circuit portion 300 shown in FIG. 2. The example shown in FIG. 14 is the same as the example shown in FIG. 10 except that the BPF 320 is disposed instead of the first BPF 322 and the second BPF 324 (FIG. 10). That is, in the example shown in FIG. 14, the signals sent from the first feed line 410 and the second feed line 420 and combined by the hybrid circuit 430 are amplified by the first-stage amplifier 312 and sent to the BPF 320. The BPF 320 passes, for example, L1 band and L band signals and L5 band, L2 band and L6 band signals. The signal extracted by the BPF 320 is amplified by the second-stage amplifier 314 and sent to the cable 510 via the attenuator 330.

In the example shown in FIGS. 8 to 13, an electrical path passing through a BPF (first BPF 322) for extracting L1 band and L band signals and an electrical path passing

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through a BPF (second BPF 324) for extracting L5 band, L2 band, and L6 band signal are split. In the example shown in FIGS. 9 to 13, the first BPF 322 and the second BPF 324 are connected in parallel. In the example shown in FIGS. 8 to 13, gains and axial ratios can be better in the L1 band, L band, L5 band, L2 band and L6 band, as compared with the case a single BPF (BPF 320) passes the L1 band and L band signals and the L5 band, L2 band, and L6 band signals as shown in FIG. 14.

In the example shown in FIGS. 8 to 14, the circuit portion 300 has an amplifier and a bandpass filter in the subsequent stage of the hybrid circuit 430 with respect to the antenna element 200. Accordingly, the circuit portion 300 can function as an LNA.

FIG. 15 is a top view of a first stacked patch antenna 910 according to comparative embodiment 1. FIG. 16 is a side view of the first stacked patch antenna 910 shown in FIG. 15.

The first stacked patch antenna 910 includes a first patch antenna 912 and a second patch antenna 914. The second patch antenna 914 is stacked on the first patch antenna 912. When viewed from above the first stacked patch antenna 910, each of the first patch antenna 912 and the second patch antenna 914 has a substantially circular shape. The size of the first stacked patch antenna 910 is 41 mm for the length L1, 41 mm for the width W1 and 13 mm for the height H1.

FIG. 17 is a perspective view of a second stacked patch antenna 920 according to the comparative embodiment 2.

The second stacked patch antenna 920 includes a third patch antenna 922 and a fourth patch antenna 924. The fourth patch antenna 924 is stacked on the third patch antenna 922. When viewed from above the second stacked patch antenna 920, the third patch antenna 922 and the fourth patch antenna 924 have a substantially square shape. The size of the first stacked patch antenna 910 is 80 mm for the length L2, 80 mm for the width W2, and 7.45 mm for the height H2.

FIG. 18 is a graph showing frequency characteristics of gain and axial ratio of the antenna element 200 (FIG. 2) according to the embodiment in the range of 1100 MHz to 1700 MHz. FIG. 19 is a graph showing frequency characteristics of gain and axial ratio of the first stacked patch antenna 910 (FIGS. 15 and 16) according to comparative embodiment 1 at 1100 MHz to 1700 MHz. FIG. 20 is a graph showing frequency characteristics of gain and axial ratio of the second stacked patch antenna 920 (FIG. 17) according to comparative embodiment 2 at 1100 MHz to 1700 MHz.

In FIGS. 18 to 20, the horizontal axis of the graph indicates frequency. The vertical axis on the left side of the graph shows the gain (dBic), and the solid line in the graph shows the frequency characteristics of the gain. The vertical axis on the right side of the graph indicates the axial ratio (dB), and the broken line in the graph indicates the frequency characteristic of the axial ratio. In the graph, the region between the thick vertical line at a frequency of about 1165 MHz and the thick vertical line at a frequency of about 1285 MHz is the L5 band, L2 band, and L6 band. In the graph, the region between the thick vertical line at a frequency of about 1525 MHz and the thick vertical line at a frequency of about 1610 MHz is the L1 band and L band.

The size of the antenna element 200 according to the embodiment is 70 mm in length (the first direction X in FIG. 2), 35 mm in width (the second direction Y in FIG. 2), and 42 mm in height (the third direction Z in FIG. 2). That is, when viewed from the height direction (third direction Z) of the antenna element 200, the space required for installing the

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antenna element **200** can be estimated as a rectangle with an area of 2450 mm^2 (70 mm (first direction X) $\times 35 \text{ mm}$ (second direction Y)). On the other hand, according to the above description, when viewed from the height direction of the first stacked patch antenna **910** according to comparative embodiment 1, the space required for installing the first stacked patch antenna **910** can be estimated as a rectangle ($41 \text{ mm}\times 41 \text{ mm}$) with an area of 1681 mm^2 . When viewed from the height direction of the second stacked patch antenna **920** according to comparative embodiment 2, the space required for installing the second stacked patch antenna **920** can be estimated as a rectangle ($80 \text{ mm}\times 80 \text{ mm}$) with an area of 6400 mm^2 .

Based on the comparison between the frequency characteristics of the first stacked patch antenna **910** according to comparative embodiment 1 (FIG. 19) and the frequency characteristics of the second stacked patch antenna **920** according to comparative embodiment 2 (FIG. 20), in order to obtain sufficient performance (gain 2.0 dBic or more and axial ratio 4.0 dB or less) in both the gain and the axial ratio in the L1 band, the L band, and the L5 band, the L2 band, and the L6 band of the stacked patch antenna, a rectangular space having an area of at least 6400 mm^2 (a space required for installing the second stacked patch antenna **920** according to comparative embodiment 2) is required when viewed from the height direction of the stacked patch antenna. On the other hand, from the frequency characteristics of the embodiment (FIG. 15), in the antenna element **200** according to the embodiment, sufficient performance (gain 2.0 dBic or more, axial ratio 4.0 dB or less) is obtained in both the gain and the axial ratio in the L1 band, the L band, and the L5 band, the L2 band, and the L6 band, with a rectangular space having an area of 2450 mm^2 when viewed from the height direction of the antenna element **200**. Thus, in the antenna element **200** according to the embodiment, sufficient performance (gain 2.0 dBic or more and an axial ratio 4.0 dB or less) can be obtained in both the gain and the axial ratio in the L1 band, the L band, the L5 band, the L2 band, and the L6 band in a space smaller than the space of the stacked patch antenna.

Although the embodiment and modification examples of the present invention have been described above with reference to the drawings, these are examples of the present invention, and various configurations other than the above may be adopted.

In the present embodiment, the first element **210** and the second element **220** of the antenna element **200** are physically supported on the mounting surface **122** of the base **100** by the first feed line **410** and the second feed line **420**. However, the first element **210** and the second element **220** of the antenna element **200** may be physically supported on the mounting surface **122** of the base **100** by an insulating block such as a resin block.

In the present embodiment, the first element **210** and the second element **220** of the antenna element **200** are made of sheet metal.

However, the first element **210** and the second element **220** of the antenna element **200** may be made of a conductive pattern patterned on an insulating block such as a resin block.

In the present embodiment, the vehicular antenna device **10** includes a base **100**, an antenna element **200**, a circuit portion **300**, a first feed line **410**, and a second feed line **420**. However, one or more other antenna elements may be provided, and for example, an antenna element for Long Term Evolution (LTE), an antenna element for Vehicle-to-Everything (V2X), and the like may be further provided.

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In the present embodiment, the first feed line **410** and the second feed line **420** are coaxial lines, but may be microstrip lines provided on the substrate.

According to the present specification, the following aspects are provided.

(Aspect 1)

Aspect 1 is a vehicular antenna device including:

an antenna element operable in at least two or more of frequency bands including L1 band, L band, L5 band, L2 band, and L6 band and receiving circular polarization, in which

the antenna element includes

a first element having a first feed portion, and a first element section and a second element section arranged at opposite sides of the first feed portion, and

a second element having a second feed portion, and a third element section and a fourth element section arranged at opposite sides of the second feed portion,

at least a part of the first element and at least a part of the second element face each other, and

one of the first element section and the second element section is arranged at an angle with respect to the other of the first element section and the second element section.

According to Aspect 1, as compared with the case where one of the first element section and the second element section is not arranged at an angle with respect to the other of the first element section and the second element section, the length of the first element in the facing direction of the first element section and the second element section can be shortened. According to Aspect 1, a part of the first element and a part of the second element face each other. Accordingly, the compatible band of the antenna element can be expanded. Thus, the vehicular antenna device compatible with a wide range of bands within a multi-band including the L1 band, the L band, the L5 band, the L2 band, and the L6 band can be miniaturized.

(Aspect 2)

Aspect 2 is the vehicular antenna device according to Aspect 1, in which

the first element section has a first portion and a second portion,

the second element section has a third portion and a fourth portion,

the third element section has a fifth portion and a sixth portion,

the fourth element section has a seventh portion and an eighth portion,

the first portion of the first element section and the fifth portion of the third element section face each other,

the second portion of the first element section and the seventh portion of the fourth element section face each other,

the third portion of the second element section and the sixth portion of the third element section face each other, and

the fourth portion of the second element section and the eighth portion of the fourth element section face each other.

According to Aspect 2, since each portion of the first element section and each portion of the second element section face each other, the compatible band of the antenna element can be expanded to the relatively low frequency band side.

(Aspect 3)

Aspect 3 is the vehicular antenna device according to Aspect 2, in which

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the first element and the second element have substantially the same shape, and the fifth portion, the sixth portion, the seventh portion, and the eighth portion are turned by about 90 degrees with respect to the third portion, the fourth portion, the first portion, and the second portion, respectively.

According to Aspect 3, the direction of polarization of the first element and the direction of polarization of the second element are orthogonal to each other. Specifically, since the first element and the second element have substantially the same shape, there is almost no difference in amplitude and phase between the linear polarization of the first element and the linear polarization of the second element, which are orthogonal to each other, and the antenna element receives circular polarization.

(Aspect 4)

Aspect 4 is the vehicular antenna device according to any one of Aspects 1 to 3, in which

the first element section and the second element section of the first element, and the third element section and the fourth element section of the second element each have a portion operating as a self-similar antenna or an equivalent antenna.

According to Aspect 4, the antenna element operates as, for example, a tapered slot antenna in a relatively high frequency band and, for example, a loop antenna in a relatively low frequency band. In a specific frequency band in an intermediate frequency band between a relatively high frequency band and a relatively low frequency band, the antenna element operates as a dipole antenna. In each band between the relatively high frequency band, the relatively low frequency band, and the intermediate frequency band, the antenna element **200** operates with the operating principles of these antennas combined, that is, a combined antenna. Accordingly, the antenna element **200** can operate stably over a wide frequency band even though it is one antenna element.

(Aspect 5)

Aspect 5 is the vehicular antenna device according to any one of Aspects 1 to 4, in which

each of the first element section, the second element section, the third element section, and the fourth element section has an opening,

the first element section and the second element section are arranged such that the opening of the first element section and the opening of the second element section face opposite each other, and

the third element section and the fourth element section are arranged such that the opening of the third element section and the opening of the fourth element section face opposite each other.

According to Aspect 5, when each of the first element section, the second element section, the third element section and the fourth element section has a portion operating as a self-similar antenna or an equivalent antenna, the antenna element can operate stably over a wide frequency band.

(Aspect 6)

Aspect 6 is the vehicular antenna device according to any one of Aspects 1 to 5, in which

each of the first element section, the second element section, the third element section, and the fourth element section is formed in any of a substantially C-shape, a substantially U-shape, a substantially V-shape, and a substantially n-shape.

According to Aspect 6, since the shape of each of the first element section, the second element section, the third ele-

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ment section and the fourth element section corresponds to a shape operating as a self-similar antenna or an equivalent antenna, the antenna element can operate stably over a wide frequency band.

5 (Aspect 7)

Aspect 7 is the vehicular antenna device according to any one of Aspects 1 to 6, further including:

an insulator provided in at least one place between the first element and the second element.

10 According to Aspect 7, the contact between the facing parts of the first element and the second element can be suppressed by the insulator. The insulator also serves as a spacer for adjusting the width of the region (gap) between the facing parts of the first element and the second element.

15 (Aspect 8)

Aspect 8 is the vehicular antenna device according to any one of Aspects 1 to 7, including:

a mounting surface on which the antenna element is mounted, in which

20 at least one of the first element section and the second element section is tilted from a direction parallel to the mounting surface toward a side on which the mounting surface is located or a side opposite to the side on which the mounting surface is located.

25 According to Aspect 8, one of the first element section and the second element section can be arranged at an angle with respect to the other of the first element section and the second element section.

(Aspect 9)

30 Aspect 9 is the vehicular antenna device according to Aspect 8, in which

the first element section and the second element section are tilted at substantially the same angle from the direction parallel to the mounting surface toward the side on which the mounting surface is located or the side opposite to the side on which the mounting surface is located.

According to Aspect 9, the radiation directivity of the antenna element in the zenith direction can be strengthened, as compared with the case where the angle of inclination of the first element section with respect to the direction parallel to the mounting surface and the angle of inclination of the second element section with respect to the direction parallel to the mounting surface are different.

45 (Aspect 10)

Aspect 10 is the vehicular antenna device according to Aspect 8, in which

the first element section and the second element section are tilted at different angles from the direction parallel to the mounting surface toward the side on which the mounting surface is located or the side opposite to the side on which the mounting surface is located.

According to Aspect 10, the radiation directivity of the antenna element can be tilted from the zenith direction toward a desired direction, by adjusting the angle of inclination of each of the first element section and the second element section with respect to the direction parallel to the mounting surface.

(Aspect 11)

60 Aspect 11 is the vehicular antenna device according to Aspect 9 or 10, in which the first element section and the second element section are tilted at an angle of more than 0 degrees and 70 degrees or less with respect to the direction parallel to the mounting surface.

65 According to Aspect 11, the decrease from the characteristics (for example, gain or axial ratio) of the antenna element with the first element section and the second ele-

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ment section disposed parallel to each other can be suppressed within a sufficiently acceptable range, and the length of the first element in the facing direction of the first element section and the second element section can be shortened. (Aspect 12)

Aspect 12 is the vehicular antenna device according to any one of Aspects 1 to 11, in which the antenna element is disposed on a ground plate.

According to Aspect 12, the antenna element can operate better as a GNSS antenna as compared to the case where the antenna element is not disposed on the ground plate. (Aspect 13)

Aspect 13 is the vehicular antenna device according to any one of Aspects 1 to 12, further including:

a hybrid circuit imparting a 90 degrees phase difference to a signal sent to the first element and a signal sent to the second element.

According to Aspect 13, the hybrid circuit allows the antenna element to receive circular polarization.

(Aspect 14)

Aspect 14 is the vehicular antenna device according to Aspect 13, further including:

a circuit portion in a subsequent stage of the hybrid circuit with respect to the antenna element, in which the circuit portion has an amplifier and a bandpass filter.

According to Aspect 14, the circuit portion can function as an LNA.

This application claims priority based on Japanese Patent Application No. 2020-011871, filed Jan. 28, 2020, the entire disclosures of which is incorporated herein.

REFERENCE SIGNS LIST

10: vehicular antenna device
 100: base
 110: first base member
 120: second base member
 122: mounting surface
 200: antenna element
 210: first element
 210a: first element section
 210b: second element section
 212a: first arm
 212b: second arm
 212c: third arm
 212d: fourth arm
 214a: first portion
 214b: second portion
 214c: third portion
 214d: fourth portion
 220: second element
 220a: third element section
 220b: fourth element section
 222a: fifth arm
 222b: sixth arm
 222c: seventh arm
 222d: eighth arm
 224a: fifth portion
 224b: sixth portion
 224c: seventh portion
 224d: eighth portion
 230: insulator
 300: circuit portion
 310: amplifier
 312: first-stage amplifier
 314: second-stage amplifier
 314a: first second-stage amplifier

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314b: second second-stage amplifier

320: BPF

322: first BPF

324: second BPF

330: attenuator

330a: first attenuator

330b: second attenuator

410: first feed line

420: second feed line

430: hybrid circuit

500: cover

510: cable

510a: first cable

510b: second cable

600: ground plate

910: first stacked patch antenna

912: first patch antenna

914: second patch antenna

920: second stacked patch antenna

922: third patch antenna

924: fourth patch antenna

X: first direction

Y: second direction

Z: third direction

The invention claimed is:

1. A vehicular antenna device comprising:

an antenna element operable in at least two or more of frequency bands including L1 band, L band, L5 band, L2 band, and L6 band and receiving circular polarization, wherein

the antenna element includes

a first element having a first feed portion, and a first element section and a second element section arranged at opposite sides of the first feed portion, and

a second element having a second feed portion, and a third element section and a fourth element section arranged at opposite sides of the second feed portion, at least a part of the first element and at least a part of the second element face each other, and

one of the first element section and the second element section is arranged at an angle with respect to the other of the first element section and the second element section,

wherein

the first element section has a first portion and a second portion,

the second element section has a third portion and a fourth portion,

the third element section has a fifth portion and a sixth portion,

the fourth element section has a seventh portion and an eighth portion,

the first portion of the first element section and the fifth portion of the third element section face each other,

the second portion of the first element section and the seventh portion of the fourth element section face each other,

the third portion of the second element section and the sixth portion of the third element section face each other, and

the fourth portion of the second element section and the eighth portion of the fourth element section face each other.

2. The vehicular antenna device according to claim 1,

wherein

the first element and the second element have substantially the same shape, and

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the fifth portion, the sixth portion, the seventh portion, and the eighth portion are turned by about 90 degrees with respect to the third portion, the fourth portion, the first portion, and the second portion, respectively.

3. The vehicular antenna device according to claim 1, 5 wherein

the first element section and the second element section of the first element, and the third element section and the fourth element section of the second element each have a portion operating as a self-similar antenna or an equivalent antenna. 10

4. The vehicular antenna device according to claim 1, wherein

each of the first element section, the second element section, the third element section, and the fourth element section has an opening. 15

the first element section and the second element section are arranged such that the opening of the first element section and the opening of the second element section face opposite each other, and 20

the third element section and the fourth element section are arranged such that the opening of the third element section and the opening of the fourth element section face opposite each other.

5. The vehicular antenna device according to claim 1, 25 wherein

each of the first element section, the second element section, the third element section, and the fourth element section is formed in any of a substantially C-shape, a substantially U-shape, a substantially V-shape, and a substantially n-shape. 30

6. The vehicular antenna device according to claim 1, further comprising:

an insulator provided in at least one place between the first element and the second element. 35

7. The vehicular antenna device according to claim 1, further comprising:

a mounting surface on which the antenna element is mounted, wherein

at least one of the first element section and the second element section is tilted from a direction parallel to the mounting surface toward a side on which the mounting surface is located or a side opposite to the side on which the mounting surface is located. 40

8. The vehicular antenna device according to claim 7, 45 wherein

the first element section and the second element section are tilted at substantially the same angle from the direction parallel to the mounting surface toward the side on which the mounting surface is located or the side opposite to the side on which the mounting surface is located. 50

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9. A vehicular antenna device comprising:

an antenna element operable in at least two or more of frequency bands including L1 band, L band, L5 band, L2 band, and L6 band and receiving circular polarization, wherein

the antenna element includes

a first element having a first feed portion, and a first element section and a second element section arranged at opposite sides of the first feed portion, and

a second element having a second feed portion, and a third element section and a fourth element section arranged at opposite sides of the second feed portion,

at least a part of the first element and at least a part of the second element face each other,

one of the first element section and the second element section is arranged at an angle with respect to the other of the first element section and the second element section; and

a mounting surface on which the antenna element is mounted, wherein

at least one of the first element section and the second element section is tilted from a direction parallel to the mounting surface toward a side on which the mounting surface is located or a side opposite to the side on which the mounting surface is located, and

the first element section and the second element section are tilted at different angles from the direction parallel to the mounting surface toward the side on which the mounting surface is located or the side opposite to the side on which the mounting surface is located.

10. The vehicular antenna device according to claim 8, wherein

the first element section and the second element section are tilted at an angle of more than 0 degrees and 70 degrees or less with respect to the direction parallel to the mounting surface.

11. The vehicular antenna device according to claim 1, wherein

the antenna element is disposed on a ground plate.

12. The vehicular antenna device according to claim 1, further comprising:

a hybrid circuit imparting a 90 degrees phase difference to a signal sent to the first element and a signal sent to the second element.

13. The vehicular antenna device according to claim 12, further comprising:

a circuit portion in a subsequent stage of the hybrid circuit with respect to the antenna element, wherein the circuit portion has an amplifier and a bandpass filter.

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